```
import os
import zipfile
import pandas as pd
import numpy as np
from nilearn.image import smooth_img
from nilearn.input_data import NiftiMasker
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout

fmri_folder = r"C:\Users\Admin\Documents\ADHD\ADHD NII"
task_folder = r"C:\Users\Admin\Documents\ADHD\ADHD TSV"

!pip install nilearn tensorflow pandas scikit-learn numpy
```

```
Requirement already satisfied: nilearn in d:\anaconda\lib\site-
packages (0.11.1)
Requirement already satisfied: tensorflow in d:\anaconda\lib\site-
packages (2.18.0)
Requirement already satisfied: pandas in d:\anaconda\lib\site-packages
(2.2.2)
Requirement already satisfied: scikit-learn in d:\anaconda\lib\site-
packages (1.5.1)
Requirement already satisfied: numpy in d:\anaconda\lib\site-packages
(1.26.4)
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(from nilearn) (5.2.1)
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Requirement already satisfied: requests>=2.25.0 in d:\anaconda\lib\
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Requirement already satisfied: tensorflow-intel==2.18.0 in d:\
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Requirement already satisfied: absl-py>=1.0.0 in d:\anaconda\lib\site-
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(25.1.24)
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```

```
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```

```
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intel==2.18.0->tensorflow) (0.1.0)
```

```
task_files = [f for f in os.listdir(task_folder) if
f.endswith(('.csv', '.tsv'))]

task_dfs = []
for file in task_files:
    file_path = os.path.join(task_folder, file)
    if file.endswith('.csv'):
        df = pd.read_csv(file_path)
    else:
        df = pd.read_csv(file_path, sep="\t")  # For TSV files
    task_dfs.append(df)

task_data = pd.concat(task_dfs, ignore_index=True)
print("Task performance data loaded successfully!")
print(task_data.head())
```

```
Task performance data loaded successfully!
    onset  duration  block block_type_intended block_type_performed  \
0    0.00     38.69      1              single               single
1   38.69     38.87      2               array                array
2   77.56     38.69      3              single               single
3  116.25     38.87      4               array                array
```

```
4  155.12     38.69      5               single                single

   response_time_avg  correct_total errors_total  false-neg  false-pos
NR  \
0             0.5679           17.0            1        NaN        1.0
NaN
1             0.6947           18.0            0        NaN        NaN
NaN
2             0.5872           18.0            0        NaN        NaN
NaN
3             0.7014           18.0            0        NaN        NaN
NaN
4             0.5265           18.0            0        NaN        NaN
NaN

  comments  false-no_go  false-go
0      NaN          NaN       NaN
1      NaN          NaN       NaN
2      NaN          NaN       NaN
3      NaN          NaN       NaN
4      NaN          NaN       NaN
```

```
import os

fmri_files = os.listdir(fmri_folder)
print("fMRI Files:", fmri_files)

task_files = os.listdir(task_folder)
print("Task Files:", task_files)
```

```
fMRI Files: ['sub-213_task-Conj19Sel_acq-a_bold.nii', 'sub-213_task-
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```
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Conj9Inh_acq-b_bold.nii', 'sub-259_task-Feat19Sel_acq-a_bold.nii',
'sub-259_task-Feat1Inh_acq-a_bold.nii', 'sub-259_task-Feat9Inh_acq-
a_bold.nii', 'sub-260_task-Conj19Sel_acq-b_bold.nii', 'sub-260_task-
Conj1Inh_acq-b_bold.nii', 'sub-260_task-Conj9Inh_acq-b_bold.nii',
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a_bold.nii', 'sub-260_task-Feat9Inh_acq-a_bold.nii', 'sub-262_task-
Conj19Sel_acq-b_bold.nii', 'sub-262_task-Conj1Inh_acq-b_bold.nii',
'sub-262_task-Conj9Inh_acq-b_bold.nii', 'sub-262_task-Feat19Sel_acq-
a_bold.nii', 'sub-262_task-Feat1Inh_acq-b_bold.nii', 'sub-262_task-
Feat9Inh_acq-a_bold.nii', 'sub-268_task-Conj19Sel_acq-b_bold.nii',
'sub-268_task-Conj1Inh_acq-b_bold.nii', 'sub-268_task-Conj9Inh_acq-
b_bold.nii', 'sub-268_task-Feat19Sel_acq-a_bold.nii', 'sub-268_task-
Feat1Inh_acq-a_bold.nii', 'sub-268_task-Feat9Inh_acq-a_bold.nii']
Task Files: ['sub-213_task-Conj19Sel_acq-a_events.tsv', 'sub-213_task-
Conj1Inh_acq-a_events.tsv', 'sub-213_task-Conj9Inh_acq-a_events.tsv',
'sub-213_task-Feat19Sel_acq-b_events.tsv', 'sub-213_task-Feat1Inh_acq-
b_events.tsv', 'sub-213_task-Feat9Inh_acq-b_events.tsv', 'sub-
214_task-Conj19Sel_acq-a_events.tsv', 'sub-214_task-Conj1Inh_acq-
a_events.tsv', 'sub-214_task-Conj9Inh_acq-a_events.tsv', 'sub-
214_task-Feat9Inh_acq-a_events.tsv', 'sub-218_task-Conj19Sel_acq-
b_events.tsv', 'sub-218_task-Conj1Inh_acq-b_events.tsv', 'sub-
218_task-Conj9Inh_acq-b_events.tsv', 'sub-218_task-Feat19Sel_acq-
a_events.tsv', 'sub-218_task-Feat1Inh_acq-a_events.tsv', 'sub-
218_task-Feat9Inh_acq-a_events.tsv', 'sub-219_task-Conj19Sel_acq-
b_events.tsv', 'sub-219_task-Conj1Inh_acq-b_events.tsv', 'sub-
219_task-Conj9Inh_acq-b_events.tsv', 'sub-219_task-Feat19Sel_acq-
a_events.tsv', 'sub-219_task-Feat1Inh_acq-a_events.tsv', 'sub-
219_task-Feat9Inh_acq-a_events.tsv', 'sub-220_task-Conj19Sel_acq-
a_events.tsv', 'sub-220_task-Conj1Inh_acq-a_events.tsv', 'sub-
220_task-Conj9Inh_acq-a_events.tsv', 'sub-220_task-Feat19Sel_acq-
b_events.tsv', 'sub-220_task-Feat1Inh_acq-b_events.tsv', 'sub-
220_task-Feat9Inh_acq-b_events.tsv', 'sub-221_task-Conj19Sel_acq-
b_events.tsv', 'sub-221_task-Conj1Inh_acq-b_events.tsv', 'sub-
221_task-Conj9Inh_acq-b_events.tsv', 'sub-221_task-Feat19Sel_acq-
a_events.tsv', 'sub-221_task-Feat1Inh_acq-a_events.tsv', 'sub-
221_task-Feat9Inh_acq-a_events.tsv', 'sub-222_task-Conj19Sel_acq-
a_events.tsv', 'sub-222_task-Conj1Inh_acq-a_events.tsv', 'sub-
222_task-Conj9Inh_acq-a_events.tsv', 'sub-222_task-Feat19Sel_acq-
a_events.tsv', 'sub-222_task-Feat1Inh_acq-a_events.tsv', 'sub-
222_task-Feat9Inh_acq-a_events.tsv', 'sub-223_task-Conj19Sel_acq-
a_events.tsv', 'sub-223_task-Conj1Inh_acq-a_events.tsv', 'sub-
223_task-Conj9Inh_acq-a_events.tsv', 'sub-223_task-Feat19Sel_acq-
a_events.tsv', 'sub-223_task-Feat1Inh_acq-a_events.tsv', 'sub-
223_task-Feat9Inh_acq-a_events.tsv', 'sub-224_task-Conj19Sel_acq-
a_events.tsv', 'sub-224_task-Conj1Inh_acq-a_events.tsv', 'sub-
224_task-Conj9Inh_acq-a_events.tsv', 'sub-224_task-Feat19Sel_acq-
```

```
a_events.tsv', 'sub-224_task-Feat1Inh_acq-a_events.tsv', 'sub-
224_task-Feat9Inh_acq-a_events.tsv', 'sub-225_task-Conj19Sel_acq-
b_events.tsv', 'sub-225_task-Conj1Inh_acq-b_events.tsv', 'sub-
225_task-Conj9Inh_acq-b_events.tsv', 'sub-225_task-Feat19Sel_acq-
a_events.tsv', 'sub-225_task-Feat1Inh_acq-a_events.tsv', 'sub-
225_task-Feat9Inh_acq-a_events.tsv', 'sub-226_task-Conj19Sel_acq-
b_events.tsv', 'sub-226_task-Conj1Inh_acq-b_events.tsv', 'sub-
226_task-Conj9Inh_acq-b_events.tsv', 'sub-226_task-Feat19Sel_acq-
a_events.tsv', 'sub-226_task-Feat1Inh_acq-a_events.tsv', 'sub-
226_task-Feat9Inh_acq-a_events.tsv', 'sub-228_task-Conj19Sel_acq-
b_events.tsv', 'sub-228_task-Conj1Inh_acq-b_events.tsv', 'sub-
228_task-Conj9Inh_acq-b_events.tsv', 'sub-228_task-Feat19Sel_acq-
a_events.tsv', 'sub-228_task-Feat1Inh_acq-a_events.tsv', 'sub-
228_task-Feat9Inh_acq-a_events.tsv', 'sub-229_task-Conj19Sel_acq-
b_events.tsv', 'sub-229_task-Conj1Inh_acq-b_events.tsv', 'sub-
229_task-Conj9Inh_acq-b_events.tsv', 'sub-229_task-Feat19Sel_acq-
a_events.tsv', 'sub-229_task-Feat1Inh_acq-a_events.tsv', 'sub-
229_task-Feat9Inh_acq-a_events.tsv', 'sub-230_task-Conj19Sel_acq-
b_events.tsv', 'sub-230_task-Conj1Inh_acq-b_events.tsv', 'sub-
230_task-Conj9Inh_acq-b_events.tsv', 'sub-230_task-Feat19Sel_acq-
a_events.tsv', 'sub-230_task-Feat1Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-231_task-Feat1Inh_acq-a_events.tsv', 'sub-
231_task-Feat9Inh_acq-a_events.tsv', 'sub-232_task-Conj19Sel_acq-
b_events.tsv', 'sub-232_task-Conj1Inh_acq-b_events.tsv', 'sub-
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a_events.tsv', 'sub-232_task-Feat1Inh_acq-a_events.tsv', 'sub-
232_task-Feat9Inh_acq-a_events.tsv', 'sub-233_task-Conj19Sel_acq-
b_events.tsv', 'sub-233_task-Conj1Inh_acq-b_events.tsv', 'sub-
233_task-Conj9Inh_acq-b_events.tsv', 'sub-233_task-Feat19Sel_acq-
a_events.tsv', 'sub-233_task-Feat1Inh_acq-a_events.tsv', 'sub-
233_task-Feat9Inh_acq-a_events.tsv', 'sub-234_task-Conj19Sel_acq-
b_events.tsv', 'sub-234_task-Conj1Inh_acq-b_events.tsv', 'sub-
234_task-Conj9Inh_acq-b_events.tsv', 'sub-234_task-Feat19Sel_acq-
a_events.tsv', 'sub-234_task-Feat1Inh_acq-a_events.tsv', 'sub-
234_task-Feat9Inh_acq-a_events.tsv', 'sub-235_task-Conj19Sel_acq-
a_events.tsv', 'sub-235_task-Conj1Inh_acq-a_events.tsv', 'sub-
235_task-Conj9Inh_acq-a_events.tsv', 'sub-235_task-Feat19Sel_acq-
a_events.tsv', 'sub-235_task-Feat1Inh_acq-a_events.tsv', 'sub-
235_task-Feat9Inh_acq-a_events.tsv', 'sub-236_task-Conj19Sel_acq-
a_events.tsv', 'sub-236_task-Conj1Inh_acq-a_events.tsv', 'sub-
236_task-Conj9Inh_acq-a_events.tsv', 'sub-236_task-Feat19Sel_acq-
b_events.tsv', 'sub-236_task-Feat1Inh_acq-b_events.tsv', 'sub-
236_task-Feat9Inh_acq-b_events.tsv', 'sub-237_task-Conj19Sel_acq-
a_events.tsv', 'sub-237_task-Conj1Inh_acq-a_events.tsv', 'sub-
237_task-Conj9Inh_acq-a_events.tsv', 'sub-237_task-Feat19Sel_acq-
a_events.tsv', 'sub-237_task-Feat1Inh_acq-a_events.tsv', 'sub-
237_task-Feat9Inh_acq-a_events.tsv', 'sub-238_task-Conj19Sel_acq-
a_events.tsv', 'sub-238_task-Conj1Inh_acq-a_events.tsv', 'sub-
```

```
238_task-Conj9Inh_acq-a_events.tsv', 'sub-238_task-Feat19Sel_acq-
a_events.tsv', 'sub-238_task-Feat1Inh_acq-a_events.tsv', 'sub-
238_task-Feat9Inh_acq-a_events.tsv', 'sub-240_task-Conj19Sel_acq-
a_events.tsv', 'sub-240_task-Conj1Inh_acq-a_events.tsv', 'sub-
240_task-Conj9Inh_acq-a_events.tsv', 'sub-240_task-Feat19Sel_acq-
b_events.tsv', 'sub-240_task-Feat1Inh_acq-b_events.tsv', 'sub-
240_task-Feat9Inh_acq-b_events.tsv', 'sub-241_task-Conj19Sel_acq-
a_events.tsv', 'sub-241_task-Conj1Inh_acq-a_events.tsv', 'sub-
241_task-Conj9Inh_acq-a_events.tsv', 'sub-241_task-Feat19Sel_acq-
a_events.tsv', 'sub-241_task-Feat1Inh_acq-a_events.tsv', 'sub-
241_task-Feat9Inh_acq-a_events.tsv', 'sub-242_task-Conj19Sel_acq-
a_events.tsv', 'sub-242_task-Conj1Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-242_task-Feat1Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-245_task-Conj1Inh_acq-b_events.tsv', 'sub-
245_task-Conj9Inh_acq-b_events.tsv', 'sub-245_task-Feat19Sel_acq-
a_events.tsv', 'sub-245_task-Feat1Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-246_task-Conj1Inh_acq-b_events.tsv', 'sub-
246_task-Conj9Inh_acq-b_events.tsv', 'sub-246_task-Feat19Sel_acq-
a_events.tsv', 'sub-246_task-Feat1Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-248_task-Conj1Inh_acq-b_events.tsv', 'sub-
248_task-Conj9Inh_acq-b_events.tsv', 'sub-248_task-Feat19Sel_acq-
a_events.tsv', 'sub-248_task-Feat1Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-250_task-Conj1Inh_acq-b_events.tsv', 'sub-
250_task-Conj9Inh_acq-b_events.tsv', 'sub-250_task-Feat19Sel_acq-
a_events.tsv', 'sub-250_task-Feat1Inh_acq-a_events.tsv', 'sub-
250_task-Feat9Inh_acq-a_events.tsv', 'sub-251_task-Conj19Sel_acq-
b_events.tsv', 'sub-251_task-Conj1Inh_acq-b_events.tsv', 'sub-
251_task-Conj9Inh_acq-b_events.tsv', 'sub-251_task-Feat19Sel_acq-
a_events.tsv', 'sub-251_task-Feat1Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-253_task-Conj1Inh_acq-a_events.tsv', 'sub-
253_task-Conj9Inh_acq-a_events.tsv', 'sub-253_task-Feat19Sel_acq-
a_events.tsv', 'sub-253_task-Feat1Inh_acq-a_events.tsv', 'sub-
253_task-Feat9Inh_acq-a_events.tsv', 'sub-254_task-Conj19Sel_acq-
b_events.tsv', 'sub-254_task-Conj1Inh_acq-b_events.tsv', 'sub-
254_task-Conj9Inh_acq-b_events.tsv', 'sub-254_task-Feat19Sel_acq-
a_events.tsv', 'sub-254_task-Feat1Inh_acq-a_events.tsv', 'sub-
254_task-Feat9Inh_acq-a_events.tsv', 'sub-256_task-Conj19Sel_acq-
a_events.tsv', 'sub-256_task-Conj1Inh_acq-a_events.tsv', 'sub-
256_task-Conj9Inh_acq-a_events.tsv', 'sub-256_task-Feat19Sel_acq-
b_events.tsv', 'sub-256_task-Feat1Inh_acq-b_events.tsv', 'sub-
256_task-Feat9Inh_acq-b_events.tsv', 'sub-258_task-Conj19Sel_acq-
b_events.tsv', 'sub-258_task-Conj1Inh_acq-b_events.tsv', 'sub-
258_task-Conj9Inh_acq-b_events.tsv', 'sub-258_task-Feat19Sel_acq-
```

```
a_events.tsv', 'sub-258_task-Feat1Inh_acq-a_events.tsv', 'sub-
258_task-Feat9Inh_acq-a_events.tsv', 'sub-259_task-Conj19Sel_acq-
b_events.tsv', 'sub-259_task-Conj1Inh_acq-b_events.tsv', 'sub-
259_task-Conj9Inh_acq-b_events.tsv', 'sub-259_task-Feat19Sel_acq-
a_events.tsv', 'sub-259_task-Feat1Inh_acq-a_events.tsv', 'sub-
259_task-Feat9Inh_acq-a_events.tsv', 'sub-260_task-Conj19Sel_acq-
b_events.tsv', 'sub-260_task-Conj1Inh_acq-b_events.tsv', 'sub-
260_task-Conj9Inh_acq-b_events.tsv', 'sub-260_task-Feat19Sel_acq-
a_events.tsv', 'sub-260_task-Feat1Inh_acq-a_events.tsv', 'sub-
260_task-Feat9Inh_acq-a_events.tsv', 'sub-262_task-Conj19Sel_acq-
b_events.tsv', 'sub-262_task-Conj1Inh_acq-b_events.tsv', 'sub-
262_task-Conj9Inh_acq-b_events.tsv', 'sub-262_task-Feat19Sel_acq-
a_events.tsv', 'sub-262_task-Feat1Inh_acq-b_events.tsv', 'sub-
262_task-Feat9Inh_acq-a_events.tsv', 'sub-268_task-Conj19Sel_acq-
b_events.tsv', 'sub-268_task-Conj1Inh_acq-b_events.tsv', 'sub-
268_task-Conj9Inh_acq-b_events.tsv', 'sub-268_task-Feat19Sel_acq-
a_events.tsv', 'sub-268_task-Feat1Inh_acq-a_events.tsv', 'sub-
268_task-Feat9Inh_acq-a_events.tsv']
```

```python
import nibabel as nib

sample_fmri_file = os.path.join(fmri_folder, fmri_files[0])
fmri_img = nib.load(sample_fmri_file)
fmri_data = fmri_img.get_fdata()

print(f"fMRI Data Shape: {fmri_data.shape}")
```

```
fMRI Data Shape: (64, 64, 32, 156)
```

```python
import pandas as pd

sample_task_file = os.path.join(task_folder, task_files[0])
task_data = pd.read_csv(sample_task_file, sep="\t")
print(task_data.head())
```

```
    onset  duration  block block_type_intended block_type_performed  \
0    0.00     38.69      1              single               single
1   38.69     38.87      2               array                array
2   77.56     38.69      3              single               single
3  116.25     38.87      4               array                array
4  155.12     38.69      5              single               single

   response_time_avg  correct_total  errors_total  false-neg  false-
pos  NR  \
0             0.5679           17.0             1        NaN
1.0 NaN
1             0.6947           18.0             0        NaN
NaN NaN
2             0.5872           18.0             0        NaN
NaN NaN
3             0.7014           18.0             0        NaN
```

```
NaN NaN
4             0.5265           18.0              0        NaN
NaN NaN

   comments
0       NaN
1       NaN
2       NaN
3       NaN
4       NaN
```

```
from nilearn.image import smooth_img

smoothed_fmri = smooth_img(sample_fmri_file, fwhm=6)
smoothed_data = smoothed_fmri.get_fdata()

print("Smoothing Applied:", smoothed_data.shape)
```

```
Smoothing Applied: (64, 64, 32, 156)
```

```
from nilearn.input_data import NiftiMasker

masker = NiftiMasker()
fmri_timeseries = masker.fit_transform(sample_fmri_file)

print(f"Extracted Time Series Shape: {fmri_timeseries.shape}")
```

```
Extracted Time Series Shape: (156, 131063)
```

```
fmri_files = [file for file in os.listdir(fmri_folder) if
file.endswith(".nii")]

from nilearn.image import smooth_img
import pandas as pd
import numpy as np
import os

def process_fmri(subject_id, fmri_path):
    smoothed_img = smooth_img(fmri_path, fwhm=6)
    fmri_features = masker.fit_transform(smoothed_img)
    fmri_features_mean = np.mean(fmri_features, axis=0)

    fmri_df = pd.DataFrame([fmri_features_mean],
                           columns=[f"voxel_{i}" for i in
range(fmri_features_mean.shape[0])])
    fmri_df["subject_id"] = subject_id
    return fmri_df

fmri_dataframes = []
for file in os.listdir(fmri_folder):
    if file.endswith(".nii"):
```

```
        subject_id = file.split("_")[0]
        fmri_path = os.path.join(fmri_folder, file)
        fmri_df = process_fmri(subject_id, fmri_path)
        fmri_dataframes.append(fmri_df)

if fmri_dataframes:
    fmri_data = pd.concat(fmri_dataframes, axis=0)
    print(" fMRI data processed successfully!")
    print(fmri_data.head())
else:
    print(" No fMRI data was processed. Check file paths.")
```

```
 fMRI data processed successfully!
    voxel_0   voxel_1   voxel_2   voxel_3   voxel_4   voxel_5
voxel_6  \
0  0.368565  0.206496  0.118393  0.093643  0.077885  0.064716
0.057494
0  0.378679  0.208021  0.124874  0.099582  0.079371  0.070807
0.055457
0  0.294402  0.155752  0.102871  0.083986  0.062104  0.047931
0.040878
0  0.325713  0.157964  0.090049  0.071630  0.060599  0.048598
0.039273
0  0.442056  0.286637  0.133547  0.086465  0.067175  0.056218
0.047615

    voxel_7   voxel_8   voxel_9  ...  voxel_131063  voxel_131064  \
0  0.048711  0.043923  0.034621  ...      0.000168      0.000250
0  0.041468  0.034994  0.031645  ...      0.000167      0.000265
0  0.036525  0.032249  0.032033  ...      0.000223      0.000325
0  0.033318  0.026330  0.021841  ...      0.000173      0.000242
0  0.040852  0.038030  0.032300  ...      0.000094      0.000163

   voxel_131065  voxel_131066  voxel_131067  voxel_131068
voxel_131069  \
0      0.000366      0.000538      0.000779      0.001230
0.003422
0      0.000376      0.000550      0.000828      0.001316
0.003727
0      0.000451      0.000636      0.000984      0.001767
0.005789
0      0.000362      0.000528      0.000803      0.001854
0.006458
0      0.000250      0.000363      0.000561      0.000936
0.002625

   voxel_131070  voxel_131071  subject_id
0      0.011862      0.021789     sub-213
0      0.012943      0.023069     sub-213
0      0.017207      0.027732     sub-213
```

```
0      0.016174      0.025112     sub-213
0      0.009052      0.016929     sub-213

[5 rows x 131073 columns]
```

```python
import os
import pandas as pd

fmri_folder = r"C:\Users\Admin\Documents\ADHD\ADHD NII"
task_folder = r"C:\Users\Admin\Documents\ADHD\ADHD TSV"

task_dataframes = []
fmri_dataframes = []

for file in os.listdir(task_folder):
    if file.endswith(".tsv"):
        subject_id = file.split("_")[0]
        task_path = os.path.join(task_folder, file)

        df = pd.read_csv(task_path, sep='\t')
        df["subject_id"] = subject_id
        task_dataframes.append(df)

task_data = pd.concat(task_dataframes, ignore_index=True)
print(" Task data processed with subject IDs:")
print(task_data.head())

for file in os.listdir(fmri_folder):
    if file.endswith(".nii.gz"):
        subject_id = file.split("_")[0]
        fmri_path = os.path.join(fmri_folder, file)

        fmri_df = process_fmri(subject_id, fmri_path)
        fmri_df["subject_id"] = subject_id
        fmri_dataframes.append(fmri_df)

fmri_data = pd.concat(fmri_dataframes, ignore_index=True)
print(" fMRI data processed with subject IDs:")
print(fmri_data.head())

final_data = pd.merge(task_data, fmri_data, on="subject_id",
how="inner")
print(" Merged dataset preview:")
print(final_data.head())

final_data.to_csv("final_task_fmri_data.csv", index=False)
print(" Data saved as 'final_task_fmri_data.csv'")
```

```
 Task data processed with subject IDs:
    onset  duration  block block_type_intended block_type_performed  \
0    0.00     38.69      1              single               single
```

```
1   38.69     38.87      2               array                array
2   77.56     38.69      3              single               single
3  116.25     38.87      4               array                array
4  155.12     38.69      5              single               single

   response_time_avg  correct_total errors_total  false-neg  false-pos
NR  \
0             0.5679           17.0            1        NaN        1.0
NaN
1             0.6947           18.0            0        NaN        NaN
NaN
2             0.5872           18.0            0        NaN        NaN
NaN
3             0.7014           18.0            0        NaN        NaN
NaN
4             0.5265           18.0            0        NaN        NaN
NaN

  comments subject_id  false-no_go  false-go
0      NaN    sub-213          NaN       NaN
1      NaN    sub-213          NaN       NaN
2      NaN    sub-213          NaN       NaN
3      NaN    sub-213          NaN       NaN
4      NaN    sub-213          NaN       NaN

---------------------------------------------------------------------
-----
ValueError                                Traceback (most recent call
last)
Cell In[28], line 37
     34         fmri_dataframes.append(fmri_df)
     36 # Combine all fMRI data
---> 37 fmri_data = pd.concat(fmri_dataframes, ignore_index=True)
     38 print("□ fMRI data processed with subject IDs:")
     39 print(fmri_data.head())

File D:\Anaconda\Lib\site-packages\pandas\core\reshape\concat.py:382,
in concat(objs, axis, join, ignore_index, keys, levels, names,
verify_integrity, sort, copy)
    379 elif copy and using_copy_on_write():
    380     copy = False
--> 382 op = _Concatenator(
    383     objs,
    384     axis=axis,
    385     ignore_index=ignore_index,
    386     join=join,
    387     keys=keys,
    388     levels=levels,
    389     names=names,
    390     verify_integrity=verify_integrity,
```

```
    391     copy=copy,
    392     sort=sort,
    393 )
    395 return op.get_result()

File D:\Anaconda\Lib\site-packages\pandas\core\reshape\concat.py:445,
in _Concatenator.__init__(self, objs, axis, join, keys, levels, names,
ignore_index, verify_integrity, copy, sort)
    442 self.verify_integrity = verify_integrity
    443 self.copy = copy
--> 445 objs, keys = self._clean_keys_and_objs(objs, keys)
    447 # figure out what our result ndim is going to be
    448 ndims = self._get_ndims(objs)

File D:\Anaconda\Lib\site-packages\pandas\core\reshape\concat.py:507,
in _Concatenator._clean_keys_and_objs(self, objs, keys)
    504     objs_list = list(objs)
    506 if len(objs_list) == 0:
--> 507     raise ValueError("No objects to concatenate")
    509 if keys is None:
    510     objs_list = list(com.not_none(*objs_list))

ValueError: No objects to concatenate
```

```
import os

fmri_folder = r"C:\Users\Admin\Documents\ADHD\ADHD NII"
task_folder = r"C:\Users\Admin\Documents\ADHD\ADHD TSV"

print(f"Checking directories:\n - fMRI: {os.path.exists(fmri_folder)}\
n - Task: {os.path.exists(task_folder)}")

print("\nTask folder contents:", os.listdir(task_folder))
print("\nfMRI folder contents:", os.listdir(fmri_folder))
```

```
Checking directories:
 - fMRI: True
 - Task: True

Task folder contents: ['sub-213_task-Conj19Sel_acq-a_events.tsv',
'sub-213_task-Conj1Inh_acq-a_events.tsv', 'sub-213_task-Conj9Inh_acq-
a_events.tsv', 'sub-213_task-Feat19Sel_acq-b_events.tsv', 'sub-
213_task-Feat1Inh_acq-b_events.tsv', 'sub-213_task-Feat9Inh_acq-
b_events.tsv', 'sub-214_task-Conj19Sel_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-214_task-Feat9Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-218_task-Conj9Inh_acq-b_events.tsv', 'sub-
218_task-Feat19Sel_acq-a_events.tsv', 'sub-218_task-Feat1Inh_acq-
a_events.tsv', 'sub-218_task-Feat9Inh_acq-a_events.tsv', 'sub-
```

```
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a_events.tsv', 'sub-219_task-Feat9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-220_task-Conj9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-222_task-Conj9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-223_task-Feat9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-224_task-Conj9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-224_task-Feat9Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-225_task-Conj9Inh_acq-b_events.tsv', 'sub-
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b_events.tsv', 'sub-226_task-Conj9Inh_acq-b_events.tsv', 'sub-
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a_events.tsv', 'sub-228_task-Feat9Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-229_task-Conj9Inh_acq-b_events.tsv', 'sub-
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a_events.tsv', 'sub-229_task-Feat9Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-230_task-Conj9Inh_acq-b_events.tsv', 'sub-
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a_events.tsv', 'sub-230_task-Feat9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-231_task-Feat9Inh_acq-a_events.tsv', 'sub-
232_task-Conj19Sel_acq-b_events.tsv', 'sub-232_task-Conj1Inh_acq-
b_events.tsv', 'sub-232_task-Conj9Inh_acq-b_events.tsv', 'sub-
232_task-Feat19Sel_acq-a_events.tsv', 'sub-232_task-Feat1Inh_acq-
```

```
a_events.tsv', 'sub-232_task-Feat9Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-233_task-Conj9Inh_acq-b_events.tsv', 'sub-
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a_events.tsv', 'sub-233_task-Feat9Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-234_task-Conj9Inh_acq-b_events.tsv', 'sub-
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a_events.tsv', 'sub-234_task-Feat9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-235_task-Conj9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-237_task-Feat9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-241_task-Conj9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-241_task-Feat9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-242_task-Conj9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-242_task-Feat9Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-245_task-Conj9Inh_acq-b_events.tsv', 'sub-
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b_events.tsv', 'sub-246_task-Conj9Inh_acq-b_events.tsv', 'sub-
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b_events.tsv', 'sub-248_task-Conj9Inh_acq-b_events.tsv', 'sub-
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```

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b_events.tsv', 'sub-254_task-Conj9Inh_acq-b_events.tsv', 'sub-
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a_events.tsv', 'sub-254_task-Feat9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-256_task-Conj9Inh_acq-a_events.tsv', 'sub-
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b_events.tsv', 'sub-256_task-Feat9Inh_acq-b_events.tsv', 'sub-
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b_events.tsv', 'sub-258_task-Conj9Inh_acq-b_events.tsv', 'sub-
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a_events.tsv', 'sub-258_task-Feat9Inh_acq-a_events.tsv', 'sub-
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a_events.tsv', 'sub-268_task-Feat9Inh_acq-a_events.tsv']

fMRI folder contents: ['sub-213_task-Conj19Sel_acq-a_bold.nii', 'sub-
213_task-Conj1Inh_acq-a_bold.nii', 'sub-213_task-Conj9Inh_acq-
a_bold.nii', 'sub-213_task-Feat19Sel_acq-b_bold.nii', 'sub-213_task-
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'sub-214_task-Conj19Sel_acq-a_bold.nii', 'sub-214_task-Conj1Inh_acq-
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```

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b_bold.nii', 'sub-218_task-Feat19Sel_acq-a_bold.nii', 'sub-218_task-
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'sub-219_task-Conj19Sel_acq-b_bold.nii', 'sub-219_task-Conj1Inh_acq-
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a_bold.nii', 'sub-226_task-Conj19Sel_acq-b_bold.nii', 'sub-226_task-
Conj1Inh_acq-b_bold.nii', 'sub-226_task-Conj9Inh_acq-b_bold.nii',
'sub-226_task-Feat19Sel_acq-a_bold.nii', 'sub-226_task-Feat1Inh_acq-
a_bold.nii', 'sub-226_task-Feat9Inh_acq-a_bold.nii', 'sub-228_task-
Conj19Sel_acq-b_bold.nii', 'sub-228_task-Conj1Inh_acq-b_bold.nii',
'sub-228_task-Conj9Inh_acq-b_bold.nii', 'sub-228_task-Feat19Sel_acq-
a_bold.nii', 'sub-228_task-Feat1Inh_acq-a_bold.nii', 'sub-228_task-
Feat9Inh_acq-a_bold.nii', 'sub-229_task-Conj19Sel_acq-b_bold.nii',
'sub-229_task-Conj1Inh_acq-b_bold.nii', 'sub-229_task-Conj9Inh_acq-
b_bold.nii', 'sub-229_task-Feat19Sel_acq-a_bold.nii', 'sub-229_task-
Feat1Inh_acq-a_bold.nii', 'sub-229_task-Feat9Inh_acq-a_bold.nii',
'sub-230_task-Conj19Sel_acq-b_bold.nii', 'sub-230_task-Conj1Inh_acq-
b_bold.nii', 'sub-230_task-Conj9Inh_acq-b_bold.nii', 'sub-230_task-
Feat19Sel_acq-a_bold.nii', 'sub-230_task-Feat1Inh_acq-a_bold.nii',
'sub-230_task-Feat9Inh_acq-a_bold.nii', 'sub-231_task-Feat19Sel_acq-
a_bold.nii', 'sub-231_task-Feat1Inh_acq-a_bold.nii', 'sub-231_task-
Feat9Inh_acq-a_bold.nii', 'sub-232_task-Conj19Sel_acq-b_bold.nii',
'sub-232_task-Conj1Inh_acq-b_bold.nii', 'sub-232_task-Conj9Inh_acq-
b_bold.nii', 'sub-232_task-Feat19Sel_acq-a_bold.nii', 'sub-232_task-
Feat1Inh_acq-a_bold.nii', 'sub-232_task-Feat9Inh_acq-a_bold.nii',
'sub-233_task-Conj19Sel_acq-b_bold.nii', 'sub-233_task-Conj1Inh_acq-
b_bold.nii', 'sub-233_task-Conj9Inh_acq-b_bold.nii', 'sub-233_task-
```

```
Feat19Sel_acq-a_bold.nii', 'sub-233_task-Feat1Inh_acq-a_bold.nii',
'sub-233_task-Feat9Inh_acq-a_bold.nii', 'sub-234_task-Conj19Sel_acq-
b_bold.nii', 'sub-234_task-Conj1Inh_acq-b_bold.nii', 'sub-234_task-
Conj9Inh_acq-b_bold.nii', 'sub-234_task-Feat19Sel_acq-a_bold.nii',
'sub-234_task-Feat1Inh_acq-a_bold.nii', 'sub-234_task-Feat9Inh_acq-
a_bold.nii', 'sub-235_task-Conj19Sel_acq-a_bold.nii', 'sub-235_task-
Conj1Inh_acq-a_bold.nii', 'sub-235_task-Conj9Inh_acq-a_bold.nii',
'sub-235_task-Feat19Sel_acq-a_bold.nii', 'sub-235_task-Feat1Inh_acq-
a_bold.nii', 'sub-235_task-Feat9Inh_acq-a_bold.nii', 'sub-236_task-
Conj19Sel_acq-a_bold.nii', 'sub-236_task-Conj1Inh_acq-a_bold.nii',
'sub-236_task-Conj9Inh_acq-a_bold.nii', 'sub-236_task-Feat19Sel_acq-
b_bold.nii', 'sub-236_task-Feat1Inh_acq-b_bold.nii', 'sub-236_task-
Feat9Inh_acq-b_bold.nii', 'sub-237_task-Conj19Sel_acq-a_bold.nii',
'sub-237_task-Conj1Inh_acq-a_bold.nii', 'sub-237_task-Conj9Inh_acq-
a_bold.nii', 'sub-237_task-Feat19Sel_acq-a_bold.nii', 'sub-237_task-
Feat1Inh_acq-a_bold.nii', 'sub-237_task-Feat9Inh_acq-a_bold.nii',
'sub-238_task-Conj19Sel_acq-a_bold.nii', 'sub-238_task-Conj1Inh_acq-
a_bold.nii', 'sub-238_task-Conj9Inh_acq-a_bold.nii', 'sub-238_task-
Feat19Sel_acq-a_bold.nii', 'sub-238_task-Feat1Inh_acq-a_bold.nii',
'sub-238_task-Feat9Inh_acq-a_bold.nii', 'sub-240_task-Conj19Sel_acq-
a_bold.nii', 'sub-240_task-Conj1Inh_acq-a_bold.nii', 'sub-240_task-
Conj9Inh_acq-a_bold.nii', 'sub-240_task-Feat19Sel_acq-b_bold.nii',
'sub-240_task-Feat1Inh_acq-b_bold.nii', 'sub-240_task-Feat9Inh_acq-
b_bold.nii', 'sub-241_task-Conj19Sel_acq-a_bold.nii', 'sub-241_task-
Conj1Inh_acq-a_bold.nii', 'sub-241_task-Conj9Inh_acq-a_bold.nii',
'sub-241_task-Feat19Sel_acq-a_bold.nii', 'sub-241_task-Feat1Inh_acq-
a_bold.nii', 'sub-241_task-Feat9Inh_acq-a_bold.nii', 'sub-242_task-
Conj19Sel_acq-a_bold.nii', 'sub-242_task-Conj1Inh_acq-a_bold.nii',
'sub-242_task-Conj9Inh_acq-a_bold.nii', 'sub-242_task-Feat19Sel_acq-
a_bold.nii', 'sub-242_task-Feat1Inh_acq-a_bold.nii', 'sub-242_task-
Feat9Inh_acq-a_bold.nii', 'sub-245_task-Conj19Sel_acq-b_bold.nii',
'sub-245_task-Conj1Inh_acq-b_bold.nii', 'sub-245_task-Conj9Inh_acq-
b_bold.nii', 'sub-245_task-Feat19Sel_acq-a_bold.nii', 'sub-245_task-
Feat1Inh_acq-a_bold.nii', 'sub-245_task-Feat9Inh_acq-a_bold.nii',
'sub-246_task-Conj19Sel_acq-b_bold.nii', 'sub-246_task-Conj1Inh_acq-
b_bold.nii', 'sub-246_task-Conj9Inh_acq-b_bold.nii', 'sub-246_task-
Feat19Sel_acq-a_bold.nii', 'sub-246_task-Feat1Inh_acq-a_bold.nii',
'sub-246_task-Feat9Inh_acq-a_bold.nii', 'sub-248_task-Conj19Sel_acq-
b_bold.nii', 'sub-248_task-Conj1Inh_acq-b_bold.nii', 'sub-248_task-
Conj9Inh_acq-b_bold.nii', 'sub-248_task-Feat19Sel_acq-a_bold.nii',
'sub-248_task-Feat1Inh_acq-a_bold.nii', 'sub-248_task-Feat9Inh_acq-
a_bold.nii', 'sub-250_task-Conj19Sel_acq-b_bold.nii', 'sub-250_task-
Conj1Inh_acq-b_bold.nii', 'sub-250_task-Conj9Inh_acq-b_bold.nii',
'sub-250_task-Feat19Sel_acq-a_bold.nii', 'sub-250_task-Feat1Inh_acq-
a_bold.nii', 'sub-250_task-Feat9Inh_acq-a_bold.nii', 'sub-251_task-
Conj19Sel_acq-b_bold.nii', 'sub-251_task-Conj1Inh_acq-b_bold.nii',
'sub-251_task-Conj9Inh_acq-b_bold.nii', 'sub-251_task-Feat19Sel_acq-
a_bold.nii', 'sub-251_task-Feat1Inh_acq-a_bold.nii', 'sub-251_task-
Feat9Inh_acq-a_bold.nii', 'sub-253_task-Conj19Sel_acq-a_bold.nii',
```

```
'sub-253_task-Conj1Inh_acq-a_bold.nii', 'sub-253_task-Conj9Inh_acq-
a_bold.nii', 'sub-253_task-Feat19Sel_acq-a_bold.nii', 'sub-253_task-
Feat1Inh_acq-a_bold.nii', 'sub-253_task-Feat9Inh_acq-a_bold.nii',
'sub-254_task-Conj19Sel_acq-b_bold.nii', 'sub-254_task-Conj1Inh_acq-
b_bold.nii', 'sub-254_task-Conj9Inh_acq-b_bold.nii', 'sub-254_task-
Feat19Sel_acq-a_bold.nii', 'sub-254_task-Feat1Inh_acq-a_bold.nii',
'sub-254_task-Feat9Inh_acq-a_bold.nii', 'sub-256_task-Conj19Sel_acq-
a_bold.nii', 'sub-256_task-Conj1Inh_acq-a_bold.nii', 'sub-256_task-
Conj9Inh_acq-a_bold.nii', 'sub-256_task-Feat19Sel_acq-b_bold.nii',
'sub-256_task-Feat1Inh_acq-b_bold.nii', 'sub-256_task-Feat9Inh_acq-
b_bold.nii', 'sub-258_task-Conj19Sel_acq-b_bold.nii', 'sub-258_task-
Conj1Inh_acq-b_bold.nii', 'sub-258_task-Conj9Inh_acq-b_bold.nii',
'sub-258_task-Feat19Sel_acq-a_bold.nii', 'sub-258_task-Feat1Inh_acq-
a_bold.nii', 'sub-258_task-Feat9Inh_acq-a_bold.nii', 'sub-259_task-
Conj19Sel_acq-b_bold.nii', 'sub-259_task-Conj1Inh_acq-b_bold.nii',
'sub-259_task-Conj9Inh_acq-b_bold.nii', 'sub-259_task-Feat19Sel_acq-
a_bold.nii', 'sub-259_task-Feat1Inh_acq-a_bold.nii', 'sub-259_task-
Feat9Inh_acq-a_bold.nii', 'sub-260_task-Conj19Sel_acq-b_bold.nii',
'sub-260_task-Conj1Inh_acq-b_bold.nii', 'sub-260_task-Conj9Inh_acq-
b_bold.nii', 'sub-260_task-Feat19Sel_acq-a_bold.nii', 'sub-260_task-
Feat1Inh_acq-a_bold.nii', 'sub-260_task-Feat9Inh_acq-a_bold.nii',
'sub-262_task-Conj19Sel_acq-b_bold.nii', 'sub-262_task-Conj1Inh_acq-
b_bold.nii', 'sub-262_task-Conj9Inh_acq-b_bold.nii', 'sub-262_task-
Feat19Sel_acq-a_bold.nii', 'sub-262_task-Feat1Inh_acq-b_bold.nii',
'sub-262_task-Feat9Inh_acq-a_bold.nii', 'sub-268_task-Conj19Sel_acq-
b_bold.nii', 'sub-268_task-Conj1Inh_acq-b_bold.nii', 'sub-268_task-
Conj9Inh_acq-b_bold.nii', 'sub-268_task-Feat19Sel_acq-a_bold.nii',
'sub-268_task-Feat1Inh_acq-a_bold.nii', 'sub-268_task-Feat9Inh_acq-
a_bold.nii']
```

```
print("\nFirst 10 task files:", os.listdir(task_folder)[:10])
print("\nFirst 10 fMRI files:", os.listdir(fmri_folder)[:10])
```

```

First 10 task files: ['sub-213_task-Conj19Sel_acq-a_events.tsv', 'sub-
213_task-Conj1Inh_acq-a_events.tsv', 'sub-213_task-Conj9Inh_acq-
a_events.tsv', 'sub-213_task-Feat19Sel_acq-b_events.tsv', 'sub-
213_task-Feat1Inh_acq-b_events.tsv', 'sub-213_task-Feat9Inh_acq-
b_events.tsv', 'sub-214_task-Conj19Sel_acq-a_events.tsv', 'sub-
214_task-Conj1Inh_acq-a_events.tsv', 'sub-214_task-Conj9Inh_acq-
a_events.tsv', 'sub-214_task-Feat9Inh_acq-a_events.tsv']

First 10 fMRI files: ['sub-213_task-Conj19Sel_acq-a_bold.nii', 'sub-
213_task-Conj1Inh_acq-a_bold.nii', 'sub-213_task-Conj9Inh_acq-
a_bold.nii', 'sub-213_task-Feat19Sel_acq-b_bold.nii', 'sub-213_task-
Feat1Inh_acq-b_bold.nii', 'sub-213_task-Feat9Inh_acq-b_bold.nii',
'sub-214_task-Conj19Sel_acq-a_bold.nii', 'sub-214_task-Conj1Inh_acq-
a_bold.nii', 'sub-214_task-Conj9Inh_acq-a_bold.nii', 'sub-214_task-
Feat9Inh_acq-a_bold.nii']
```

```python
import os
import pandas as pd

def extract_subject_id(filename):
    return filename.split("_")[0]

task_dataframes = []
for file in os.listdir(task_folder):
    if file.endswith(".tsv"):
        file_path = os.path.join(task_folder, file)
        df = pd.read_csv(file_path, sep="\t")
        df["subject_id"] = extract_subject_id(file)
        task_dataframes.append(df)

task_data = pd.concat(task_dataframes, ignore_index=True)

fmri_dataframes = []
for file in os.listdir(fmri_folder):
    if file.endswith(".nii.gz"):
        subject_id = extract_subject_id(file)
        fmri_dataframes.append({"subject_id": subject_id})
fmri_data = pd.DataFrame(fmri_dataframes)


print("Task Data Sample:\n", task_data.head())
print("\nFMRI Data Sample:\n", fmri_data.head())
```

```
Task Data Sample:
     onset  duration  block block_type_intended
block_type_performed  \
0    0.00     38.69      1              single                single
1   38.69     38.87      2               array                 array
2   77.56     38.69      3              single                single
3  116.25     38.87      4               array                 array
4  155.12     38.69      5              single                single

   response_time_avg  correct_total errors_total  false-neg  false-pos
NR  \
0             0.5679           17.0            1        NaN        1.0
NaN
1             0.6947           18.0            0        NaN        NaN
NaN
2             0.5872           18.0            0        NaN        NaN
NaN
3             0.7014           18.0            0        NaN        NaN
NaN
4             0.5265           18.0            0        NaN        NaN
NaN

  comments subject_id  false-no_go  false-go
```

```
0      NaN    sub-213          NaN       NaN
1      NaN    sub-213          NaN       NaN
2      NaN    sub-213          NaN       NaN
3      NaN    sub-213          NaN       NaN
4      NaN    sub-213          NaN       NaN

FMRI Data Sample:
 Empty DataFrame
Columns: []
Index: []
```

```
import os

for file in os.listdir(task_folder):
    if file.endswith(".tsv"):
        print(f"Task File: {file} → Extracted Subject ID:
{file.split('_')[0]}")

for file in os.listdir(fmri_folder):
    if file.endswith(".nii.gz"):
        print(f"fMRI File: {file} → Extracted Subject ID:
{file.split('_')[0]}")
```

```
Task File: sub-213_task-Conj19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-213
Task File: sub-213_task-Conj1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-213
Task File: sub-213_task-Conj9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-213
Task File: sub-213_task-Feat19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-213
Task File: sub-213_task-Feat1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-213
Task File: sub-213_task-Feat9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-213
Task File: sub-214_task-Conj19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-214
Task File: sub-214_task-Conj1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-214
Task File: sub-214_task-Conj9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-214
Task File: sub-214_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-214
Task File: sub-218_task-Conj19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-218
Task File: sub-218_task-Conj1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-218
Task File: sub-218_task-Conj9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-218
Task File: sub-218_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
```

```
ID: sub-218
Task File: sub-218_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-218
Task File: sub-218_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-218
Task File: sub-219_task-Conj19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-219
Task File: sub-219_task-Conj1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-219
Task File: sub-219_task-Conj9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-219
Task File: sub-219_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-219
Task File: sub-219_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-219
Task File: sub-219_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-219
Task File: sub-220_task-Conj19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-220
Task File: sub-220_task-Conj1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-220
Task File: sub-220_task-Conj9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-220
Task File: sub-220_task-Feat19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-220
Task File: sub-220_task-Feat1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-220
Task File: sub-220_task-Feat9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-220
Task File: sub-221_task-Conj19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-221
Task File: sub-221_task-Conj1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-221
Task File: sub-221_task-Conj9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-221
Task File: sub-221_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-221
Task File: sub-221_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-221
Task File: sub-221_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-221
Task File: sub-222_task-Conj19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-222
Task File: sub-222_task-Conj1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-222
Task File: sub-222_task-Conj9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-222
Task File: sub-222_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-222
```

```
Task File: sub-222_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-222
Task File: sub-222_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-222
Task File: sub-223_task-Conj19Sel_acq-a_events.tsv → Extracted Subject ID: sub-223
Task File: sub-223_task-Conj1Inh_acq-a_events.tsv → Extracted Subject ID: sub-223
Task File: sub-223_task-Conj9Inh_acq-a_events.tsv → Extracted Subject ID: sub-223
Task File: sub-223_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-223
Task File: sub-223_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-223
Task File: sub-223_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-223
Task File: sub-224_task-Conj19Sel_acq-a_events.tsv → Extracted Subject ID: sub-224
Task File: sub-224_task-Conj1Inh_acq-a_events.tsv → Extracted Subject ID: sub-224
Task File: sub-224_task-Conj9Inh_acq-a_events.tsv → Extracted Subject ID: sub-224
Task File: sub-224_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-224
Task File: sub-224_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-224
Task File: sub-224_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-224
Task File: sub-225_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-225
Task File: sub-225_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-225
Task File: sub-225_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-225
Task File: sub-225_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-225
Task File: sub-225_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-225
Task File: sub-225_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-225
Task File: sub-226_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-226
Task File: sub-226_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-226
Task File: sub-226_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-226
Task File: sub-226_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-226
Task File: sub-226_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
```

```
ID: sub-226
Task File: sub-226_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-226
Task File: sub-228_task-Conj19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-228
Task File: sub-228_task-Conj1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-228
Task File: sub-228_task-Conj9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-228
Task File: sub-228_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-228
Task File: sub-228_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-228
Task File: sub-228_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-228
Task File: sub-229_task-Conj19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-229
Task File: sub-229_task-Conj1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-229
Task File: sub-229_task-Conj9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-229
Task File: sub-229_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-229
Task File: sub-229_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-229
Task File: sub-229_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-229
Task File: sub-230_task-Conj19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-230
Task File: sub-230_task-Conj1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-230
Task File: sub-230_task-Conj9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-230
Task File: sub-230_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-230
Task File: sub-230_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-230
Task File: sub-230_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-230
Task File: sub-231_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-231
Task File: sub-231_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-231
Task File: sub-231_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-231
Task File: sub-232_task-Conj19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-232
Task File: sub-232_task-Conj1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-232
```

```
Task File: sub-232_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-232
Task File: sub-232_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-232
Task File: sub-232_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-232
Task File: sub-232_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-232
Task File: sub-233_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-233
Task File: sub-233_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-233
Task File: sub-233_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-233
Task File: sub-233_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-233
Task File: sub-233_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-233
Task File: sub-233_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-233
Task File: sub-234_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-234
Task File: sub-234_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-234
Task File: sub-234_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-234
Task File: sub-234_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-234
Task File: sub-234_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-234
Task File: sub-234_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-234
Task File: sub-235_task-Conj19Sel_acq-a_events.tsv → Extracted Subject ID: sub-235
Task File: sub-235_task-Conj1Inh_acq-a_events.tsv → Extracted Subject ID: sub-235
Task File: sub-235_task-Conj9Inh_acq-a_events.tsv → Extracted Subject ID: sub-235
Task File: sub-235_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-235
Task File: sub-235_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-235
Task File: sub-235_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-235
Task File: sub-236_task-Conj19Sel_acq-a_events.tsv → Extracted Subject ID: sub-236
Task File: sub-236_task-Conj1Inh_acq-a_events.tsv → Extracted Subject ID: sub-236
Task File: sub-236_task-Conj9Inh_acq-a_events.tsv → Extracted Subject
```

```
ID: sub-236
Task File: sub-236_task-Feat19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-236
Task File: sub-236_task-Feat1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-236
Task File: sub-236_task-Feat9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-236
Task File: sub-237_task-Conj19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-237
Task File: sub-237_task-Conj1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-237
Task File: sub-237_task-Conj9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-237
Task File: sub-237_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-237
Task File: sub-237_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-237
Task File: sub-237_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-237
Task File: sub-238_task-Conj19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-238
Task File: sub-238_task-Conj1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-238
Task File: sub-238_task-Conj9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-238
Task File: sub-238_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-238
Task File: sub-238_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-238
Task File: sub-238_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-238
Task File: sub-240_task-Conj19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-240
Task File: sub-240_task-Conj1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-240
Task File: sub-240_task-Conj9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-240
Task File: sub-240_task-Feat19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-240
Task File: sub-240_task-Feat1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-240
Task File: sub-240_task-Feat9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-240
Task File: sub-241_task-Conj19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-241
Task File: sub-241_task-Conj1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-241
Task File: sub-241_task-Conj9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-241
```

```
Task File: sub-241_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-241
Task File: sub-241_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-241
Task File: sub-241_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-241
Task File: sub-242_task-Conj19Sel_acq-a_events.tsv → Extracted Subject ID: sub-242
Task File: sub-242_task-Conj1Inh_acq-a_events.tsv → Extracted Subject ID: sub-242
Task File: sub-242_task-Conj9Inh_acq-a_events.tsv → Extracted Subject ID: sub-242
Task File: sub-242_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-242
Task File: sub-242_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-242
Task File: sub-242_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-242
Task File: sub-245_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-245
Task File: sub-245_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-245
Task File: sub-245_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-245
Task File: sub-245_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-245
Task File: sub-245_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-245
Task File: sub-245_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-245
Task File: sub-246_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-246
Task File: sub-246_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-246
Task File: sub-246_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-246
Task File: sub-246_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-246
Task File: sub-246_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-246
Task File: sub-246_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-246
Task File: sub-248_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-248
Task File: sub-248_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-248
Task File: sub-248_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-248
Task File: sub-248_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
```

```
ID: sub-248
Task File: sub-248_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-248
Task File: sub-248_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-248
Task File: sub-250_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-250
Task File: sub-250_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-250
Task File: sub-250_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-250
Task File: sub-250_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-250
Task File: sub-250_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-250
Task File: sub-250_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-250
Task File: sub-251_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-251
Task File: sub-251_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-251
Task File: sub-251_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-251
Task File: sub-251_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-251
Task File: sub-251_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-251
Task File: sub-251_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-251
Task File: sub-253_task-Conj19Sel_acq-a_events.tsv → Extracted Subject ID: sub-253
Task File: sub-253_task-Conj1Inh_acq-a_events.tsv → Extracted Subject ID: sub-253
Task File: sub-253_task-Conj9Inh_acq-a_events.tsv → Extracted Subject ID: sub-253
Task File: sub-253_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-253
Task File: sub-253_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-253
Task File: sub-253_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-253
Task File: sub-254_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-254
Task File: sub-254_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-254
Task File: sub-254_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-254
Task File: sub-254_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-254
```

```
Task File: sub-254_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-254
Task File: sub-254_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-254
Task File: sub-256_task-Conj19Sel_acq-a_events.tsv → Extracted Subject ID: sub-256
Task File: sub-256_task-Conj1Inh_acq-a_events.tsv → Extracted Subject ID: sub-256
Task File: sub-256_task-Conj9Inh_acq-a_events.tsv → Extracted Subject ID: sub-256
Task File: sub-256_task-Feat19Sel_acq-b_events.tsv → Extracted Subject ID: sub-256
Task File: sub-256_task-Feat1Inh_acq-b_events.tsv → Extracted Subject ID: sub-256
Task File: sub-256_task-Feat9Inh_acq-b_events.tsv → Extracted Subject ID: sub-256
Task File: sub-258_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-258
Task File: sub-258_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-258
Task File: sub-258_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-258
Task File: sub-258_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-258
Task File: sub-258_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-258
Task File: sub-258_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-258
Task File: sub-259_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-259
Task File: sub-259_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-259
Task File: sub-259_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-259
Task File: sub-259_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-259
Task File: sub-259_task-Feat1Inh_acq-a_events.tsv → Extracted Subject ID: sub-259
Task File: sub-259_task-Feat9Inh_acq-a_events.tsv → Extracted Subject ID: sub-259
Task File: sub-260_task-Conj19Sel_acq-b_events.tsv → Extracted Subject ID: sub-260
Task File: sub-260_task-Conj1Inh_acq-b_events.tsv → Extracted Subject ID: sub-260
Task File: sub-260_task-Conj9Inh_acq-b_events.tsv → Extracted Subject ID: sub-260
Task File: sub-260_task-Feat19Sel_acq-a_events.tsv → Extracted Subject ID: sub-260
Task File: sub-260_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
```

```
ID: sub-260
Task File: sub-260_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-260
Task File: sub-262_task-Conj19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-262
Task File: sub-262_task-Conj1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-262
Task File: sub-262_task-Conj9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-262
Task File: sub-262_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-262
Task File: sub-262_task-Feat1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-262
Task File: sub-262_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-262
Task File: sub-268_task-Conj19Sel_acq-b_events.tsv → Extracted Subject
ID: sub-268
Task File: sub-268_task-Conj1Inh_acq-b_events.tsv → Extracted Subject
ID: sub-268
Task File: sub-268_task-Conj9Inh_acq-b_events.tsv → Extracted Subject
ID: sub-268
Task File: sub-268_task-Feat19Sel_acq-a_events.tsv → Extracted Subject
ID: sub-268
Task File: sub-268_task-Feat1Inh_acq-a_events.tsv → Extracted Subject
ID: sub-268
Task File: sub-268_task-Feat9Inh_acq-a_events.tsv → Extracted Subject
ID: sub-268
```

```
task_dataframes = []
for file in os.listdir(task_folder):
    if file.endswith(".tsv"):
        file_path = os.path.join(task_folder, file)
        df = pd.read_csv(file_path, sep="\t")
        df["subject_id"] = file.split("_")[0]
        task_dataframes.append(df)

task_data = pd.concat(task_dataframes, ignore_index=True)

print("Task Data Sample with Subject ID:\n", task_data.head())
```

```
Task Data Sample with Subject ID:
     onset  duration  block block_type_intended
block_type_performed  \
0    0.00     38.69      1              single               single
1   38.69     38.87      2               array                array
2   77.56     38.69      3              single               single
3  116.25     38.87      4               array                array
4  155.12     38.69      5              single               single

   response_time_avg  correct_total errors_total  false-neg  false-pos
```

```
NR  \
0              0.5679            17.0             1         NaN         1.0
NaN
1              0.6947            18.0             0         NaN         NaN
NaN
2              0.5872            18.0             0         NaN         NaN
NaN
3              0.7014            18.0             0         NaN         NaN
NaN
4              0.5265            18.0             0         NaN         NaN
NaN

  comments subject_id  false-no_go  false-go
0      NaN    sub-213          NaN       NaN
1      NaN    sub-213          NaN       NaN
2      NaN    sub-213          NaN       NaN
3      NaN    sub-213          NaN       NaN
4      NaN    sub-213          NaN       NaN
```

```
fmri_dataframes = []
for file in os.listdir(fmri_folder):
    if file.endswith(".nii.gz"):
        subject_id = file.split("_")[0]
        fmri_dataframes.append({"subject_id": subject_id})

fmri_data = pd.DataFrame(fmri_dataframes)

print("fMRI Data Sample with Subject ID:\n", fmri_data.head())
```

```
fMRI Data Sample with Subject ID:
 Empty DataFrame
Columns: []
Index: []
```

```
import os

fmri_dataframes = []
for file in os.listdir(fmri_folder):
    print(f"Checking File: {file}")
    if file.endswith(".nii.gz"):
        subject_id = file.split("_")[0]
        print(f"Extracted Subject ID: {subject_id}")
        fmri_dataframes.append({"subject_id": subject_id})

fmri_data = pd.DataFrame(fmri_dataframes)

print("Final fMRI Data:\n", fmri_data)
```

```
Checking File: sub-213_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-213_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-213_task-Conj9Inh_acq-a_bold.nii
```

```
Checking File: sub-213_task-Feat19Sel_acq-b_bold.nii
Checking File: sub-213_task-Feat1Inh_acq-b_bold.nii
Checking File: sub-213_task-Feat9Inh_acq-b_bold.nii
Checking File: sub-214_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-214_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-214_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-214_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-218_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-218_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-218_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-218_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-218_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-218_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-219_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-219_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-219_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-219_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-219_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-219_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-220_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-220_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-220_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-220_task-Feat19Sel_acq-b_bold.nii
Checking File: sub-220_task-Feat1Inh_acq-b_bold.nii
Checking File: sub-220_task-Feat9Inh_acq-b_bold.nii
Checking File: sub-221_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-221_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-221_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-221_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-221_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-221_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-222_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-222_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-222_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-222_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-222_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-222_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-223_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-223_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-223_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-223_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-223_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-223_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-224_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-224_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-224_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-224_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-224_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-224_task-Feat9Inh_acq-a_bold.nii
```

```
Checking File: sub-225_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-225_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-225_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-225_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-225_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-225_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-226_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-226_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-226_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-226_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-226_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-226_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-228_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-228_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-228_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-228_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-228_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-228_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-229_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-229_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-229_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-229_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-229_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-229_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-230_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-230_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-230_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-230_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-230_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-230_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-231_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-231_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-231_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-232_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-232_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-232_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-232_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-232_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-232_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-233_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-233_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-233_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-233_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-233_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-233_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-234_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-234_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-234_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-234_task-Feat19Sel_acq-a_bold.nii
```

```
Checking File: sub-234_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-234_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-235_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-235_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-235_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-235_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-235_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-235_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-236_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-236_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-236_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-236_task-Feat19Sel_acq-b_bold.nii
Checking File: sub-236_task-Feat1Inh_acq-b_bold.nii
Checking File: sub-236_task-Feat9Inh_acq-b_bold.nii
Checking File: sub-237_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-237_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-237_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-237_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-237_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-237_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-238_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-238_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-238_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-238_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-238_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-238_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-240_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-240_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-240_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-240_task-Feat19Sel_acq-b_bold.nii
Checking File: sub-240_task-Feat1Inh_acq-b_bold.nii
Checking File: sub-240_task-Feat9Inh_acq-b_bold.nii
Checking File: sub-241_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-241_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-241_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-241_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-241_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-241_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-242_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-242_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-242_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-242_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-242_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-242_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-245_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-245_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-245_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-245_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-245_task-Feat1Inh_acq-a_bold.nii
```

```
Checking File: sub-245_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-246_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-246_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-246_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-246_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-246_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-246_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-248_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-248_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-248_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-248_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-248_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-248_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-250_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-250_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-250_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-250_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-250_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-250_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-251_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-251_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-251_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-251_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-251_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-251_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-253_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-253_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-253_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-253_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-253_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-253_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-254_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-254_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-254_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-254_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-254_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-254_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-256_task-Conj19Sel_acq-a_bold.nii
Checking File: sub-256_task-Conj1Inh_acq-a_bold.nii
Checking File: sub-256_task-Conj9Inh_acq-a_bold.nii
Checking File: sub-256_task-Feat19Sel_acq-b_bold.nii
Checking File: sub-256_task-Feat1Inh_acq-b_bold.nii
Checking File: sub-256_task-Feat9Inh_acq-b_bold.nii
Checking File: sub-258_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-258_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-258_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-258_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-258_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-258_task-Feat9Inh_acq-a_bold.nii
```

```
Checking File: sub-259_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-259_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-259_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-259_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-259_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-259_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-260_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-260_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-260_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-260_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-260_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-260_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-262_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-262_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-262_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-262_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-262_task-Feat1Inh_acq-b_bold.nii
Checking File: sub-262_task-Feat9Inh_acq-a_bold.nii
Checking File: sub-268_task-Conj19Sel_acq-b_bold.nii
Checking File: sub-268_task-Conj1Inh_acq-b_bold.nii
Checking File: sub-268_task-Conj9Inh_acq-b_bold.nii
Checking File: sub-268_task-Feat19Sel_acq-a_bold.nii
Checking File: sub-268_task-Feat1Inh_acq-a_bold.nii
Checking File: sub-268_task-Feat9Inh_acq-a_bold.nii
Final fMRI Data:
 Empty DataFrame
Columns: []
Index: []
```

```
print([file for file in os.listdir(fmri_folder)])
```

```
['sub-213_task-Conj19Sel_acq-a_bold.nii', 'sub-213_task-Conj1Inh_acq-
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Feat19Sel_acq-b_bold.nii', 'sub-213_task-Feat1Inh_acq-b_bold.nii',
'sub-213_task-Feat9Inh_acq-b_bold.nii', 'sub-214_task-Conj19Sel_acq-
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'sub-218_task-Conj19Sel_acq-b_bold.nii', 'sub-218_task-Conj1Inh_acq-
b_bold.nii', 'sub-218_task-Conj9Inh_acq-b_bold.nii', 'sub-218_task-
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'sub-218_task-Feat9Inh_acq-a_bold.nii', 'sub-219_task-Conj19Sel_acq-
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'sub-219_task-Feat1Inh_acq-a_bold.nii', 'sub-219_task-Feat9Inh_acq-
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'sub-221_task-Conj9Inh_acq-b_bold.nii', 'sub-221_task-Feat19Sel_acq-
```

```
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'sub-222_task-Conj1Inh_acq-a_bold.nii', 'sub-222_task-Conj9Inh_acq-
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'sub-228_task-Conj1Inh_acq-b_bold.nii', 'sub-228_task-Conj9Inh_acq-
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'sub-230_task-Feat1Inh_acq-a_bold.nii', 'sub-230_task-Feat9Inh_acq-
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a_bold.nii', 'sub-234_task-Feat9Inh_acq-a_bold.nii', 'sub-235_task-
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Feat9Inh_acq-a_bold.nii', 'sub-236_task-Conj19Sel_acq-a_bold.nii',
'sub-236_task-Conj1Inh_acq-a_bold.nii', 'sub-236_task-Conj9Inh_acq-
a_bold.nii', 'sub-236_task-Feat19Sel_acq-b_bold.nii', 'sub-236_task-
```

```
Feat1Inh_acq-b_bold.nii', 'sub-236_task-Feat9Inh_acq-b_bold.nii',
'sub-237_task-Conj19Sel_acq-a_bold.nii', 'sub-237_task-Conj1Inh_acq-
a_bold.nii', 'sub-237_task-Conj9Inh_acq-a_bold.nii', 'sub-237_task-
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a_bold.nii', 'sub-240_task-Conj19Sel_acq-a_bold.nii', 'sub-240_task-
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'sub-253_task-Conj19Sel_acq-a_bold.nii', 'sub-253_task-Conj1Inh_acq-
a_bold.nii', 'sub-253_task-Conj9Inh_acq-a_bold.nii', 'sub-253_task-
Feat19Sel_acq-a_bold.nii', 'sub-253_task-Feat1Inh_acq-a_bold.nii',
'sub-253_task-Feat9Inh_acq-a_bold.nii', 'sub-254_task-Conj19Sel_acq-
b_bold.nii', 'sub-254_task-Conj1Inh_acq-b_bold.nii', 'sub-254_task-
Conj9Inh_acq-b_bold.nii', 'sub-254_task-Feat19Sel_acq-a_bold.nii',
'sub-254_task-Feat1Inh_acq-a_bold.nii', 'sub-254_task-Feat9Inh_acq-
a_bold.nii', 'sub-256_task-Conj19Sel_acq-a_bold.nii', 'sub-256_task-
Conj1Inh_acq-a_bold.nii', 'sub-256_task-Conj9Inh_acq-a_bold.nii',
'sub-256_task-Feat19Sel_acq-b_bold.nii', 'sub-256_task-Feat1Inh_acq-
b_bold.nii', 'sub-256_task-Feat9Inh_acq-b_bold.nii', 'sub-258_task-
Conj19Sel_acq-b_bold.nii', 'sub-258_task-Conj1Inh_acq-b_bold.nii',
```

```
'sub-258_task-Conj9Inh_acq-b_bold.nii', 'sub-258_task-Feat19Sel_acq-a_bold.nii', 'sub-258_task-Feat1Inh_acq-a_bold.nii', 'sub-258_task-Feat9Inh_acq-a_bold.nii', 'sub-259_task-Conj19Sel_acq-b_bold.nii', 'sub-259_task-Conj1Inh_acq-b_bold.nii', 'sub-259_task-Conj9Inh_acq-b_bold.nii', 'sub-259_task-Feat19Sel_acq-a_bold.nii', 'sub-259_task-Feat1Inh_acq-a_bold.nii', 'sub-259_task-Feat9Inh_acq-a_bold.nii', 'sub-260_task-Conj19Sel_acq-b_bold.nii', 'sub-260_task-Conj1Inh_acq-b_bold.nii', 'sub-260_task-Conj9Inh_acq-b_bold.nii', 'sub-260_task-Feat19Sel_acq-a_bold.nii', 'sub-260_task-Feat1Inh_acq-a_bold.nii', 'sub-260_task-Feat9Inh_acq-a_bold.nii', 'sub-262_task-Conj19Sel_acq-b_bold.nii', 'sub-262_task-Conj1Inh_acq-b_bold.nii', 'sub-262_task-Conj9Inh_acq-b_bold.nii', 'sub-262_task-Feat19Sel_acq-a_bold.nii', 'sub-262_task-Feat1Inh_acq-b_bold.nii', 'sub-262_task-Feat9Inh_acq-a_bold.nii', 'sub-268_task-Conj19Sel_acq-b_bold.nii', 'sub-268_task-Conj1Inh_acq-b_bold.nii', 'sub-268_task-Conj9Inh_acq-b_bold.nii', 'sub-268_task-Feat19Sel_acq-a_bold.nii', 'sub-268_task-Feat1Inh_acq-a_bold.nii', 'sub-268_task-Feat9Inh_acq-a_bold.nii']
```

```
fmri_dataframes = []
for file in os.listdir(fmri_folder):
    print(f"Checking File: {file}")
    if file.endswith(".nii"):
        subject_id = file.split("_")[0]
        print(f"Extracted Subject ID: {subject_id}")
        fmri_dataframes.append({"subject_id": subject_id})

fmri_data = pd.DataFrame(fmri_dataframes)

print("Final fMRI Data:\n", fmri_data)
```

```
Checking File: sub-213_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-213
Checking File: sub-213_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-213
Checking File: sub-213_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-213
Checking File: sub-213_task-Feat19Sel_acq-b_bold.nii
Extracted Subject ID: sub-213
Checking File: sub-213_task-Feat1Inh_acq-b_bold.nii
Extracted Subject ID: sub-213
Checking File: sub-213_task-Feat9Inh_acq-b_bold.nii
Extracted Subject ID: sub-213
Checking File: sub-214_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-214
Checking File: sub-214_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-214
Checking File: sub-214_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-214
Checking File: sub-214_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-214
```

```
Checking File: sub-218_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-218
Checking File: sub-218_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-218
Checking File: sub-218_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-218
Checking File: sub-218_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-218
Checking File: sub-218_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-218
Checking File: sub-218_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-218
Checking File: sub-219_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-219
Checking File: sub-219_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-219
Checking File: sub-219_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-219
Checking File: sub-219_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-219
Checking File: sub-219_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-219
Checking File: sub-219_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-219
Checking File: sub-220_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-220
Checking File: sub-220_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-220
Checking File: sub-220_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-220
Checking File: sub-220_task-Feat19Sel_acq-b_bold.nii
Extracted Subject ID: sub-220
Checking File: sub-220_task-Feat1Inh_acq-b_bold.nii
Extracted Subject ID: sub-220
Checking File: sub-220_task-Feat9Inh_acq-b_bold.nii
Extracted Subject ID: sub-220
Checking File: sub-221_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-221
Checking File: sub-221_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-221
Checking File: sub-221_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-221
Checking File: sub-221_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-221
Checking File: sub-221_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-221
Checking File: sub-221_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-221
Checking File: sub-222_task-Conj19Sel_acq-a_bold.nii
```

```
Extracted Subject ID: sub-222
Checking File: sub-222_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-222
Checking File: sub-222_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-222
Checking File: sub-222_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-222
Checking File: sub-222_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-222
Checking File: sub-222_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-222
Checking File: sub-223_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-223
Checking File: sub-223_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-223
Checking File: sub-223_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-223
Checking File: sub-223_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-223
Checking File: sub-223_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-223
Checking File: sub-223_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-223
Checking File: sub-224_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-224
Checking File: sub-224_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-224
Checking File: sub-224_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-224
Checking File: sub-224_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-224
Checking File: sub-224_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-224
Checking File: sub-224_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-224
Checking File: sub-225_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-225
Checking File: sub-225_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-225
Checking File: sub-225_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-225
Checking File: sub-225_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-225
Checking File: sub-225_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-225
Checking File: sub-225_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-225
Checking File: sub-226_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-226
```

```
Checking File: sub-226_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-226
Checking File: sub-226_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-226
Checking File: sub-226_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-226
Checking File: sub-226_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-226
Checking File: sub-226_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-226
Checking File: sub-228_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-228
Checking File: sub-228_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-228
Checking File: sub-228_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-228
Checking File: sub-228_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-228
Checking File: sub-228_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-228
Checking File: sub-228_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-228
Checking File: sub-229_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-229
Checking File: sub-229_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-229
Checking File: sub-229_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-229
Checking File: sub-229_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-229
Checking File: sub-229_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-229
Checking File: sub-229_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-229
Checking File: sub-230_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-230
Checking File: sub-230_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-230
Checking File: sub-230_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-230
Checking File: sub-230_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-230
Checking File: sub-230_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-230
Checking File: sub-230_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-230
Checking File: sub-231_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-231
Checking File: sub-231_task-Feat1Inh_acq-a_bold.nii
```

```
Extracted Subject ID: sub-231
Checking File: sub-231_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-231
Checking File: sub-232_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-232
Checking File: sub-232_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-232
Checking File: sub-232_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-232
Checking File: sub-232_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-232
Checking File: sub-232_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-232
Checking File: sub-232_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-232
Checking File: sub-233_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-233
Checking File: sub-233_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-233
Checking File: sub-233_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-233
Checking File: sub-233_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-233
Checking File: sub-233_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-233
Checking File: sub-233_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-233
Checking File: sub-234_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-234
Checking File: sub-234_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-234
Checking File: sub-234_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-234
Checking File: sub-234_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-234
Checking File: sub-234_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-234
Checking File: sub-234_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-234
Checking File: sub-235_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-235
Checking File: sub-235_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-235
Checking File: sub-235_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-235
Checking File: sub-235_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-235
Checking File: sub-235_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-235
```

```
Checking File: sub-235_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-235
Checking File: sub-236_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-236
Checking File: sub-236_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-236
Checking File: sub-236_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-236
Checking File: sub-236_task-Feat19Sel_acq-b_bold.nii
Extracted Subject ID: sub-236
Checking File: sub-236_task-Feat1Inh_acq-b_bold.nii
Extracted Subject ID: sub-236
Checking File: sub-236_task-Feat9Inh_acq-b_bold.nii
Extracted Subject ID: sub-236
Checking File: sub-237_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-237
Checking File: sub-237_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-237
Checking File: sub-237_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-237
Checking File: sub-237_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-237
Checking File: sub-237_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-237
Checking File: sub-237_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-237
Checking File: sub-238_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-238
Checking File: sub-238_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-238
Checking File: sub-238_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-238
Checking File: sub-238_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-238
Checking File: sub-238_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-238
Checking File: sub-238_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-238
Checking File: sub-240_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-240
Checking File: sub-240_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-240
Checking File: sub-240_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-240
Checking File: sub-240_task-Feat19Sel_acq-b_bold.nii
Extracted Subject ID: sub-240
Checking File: sub-240_task-Feat1Inh_acq-b_bold.nii
Extracted Subject ID: sub-240
Checking File: sub-240_task-Feat9Inh_acq-b_bold.nii
```

```
Extracted Subject ID: sub-240
Checking File: sub-241_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-241
Checking File: sub-241_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-241
Checking File: sub-241_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-241
Checking File: sub-241_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-241
Checking File: sub-241_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-241
Checking File: sub-241_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-241
Checking File: sub-242_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-242
Checking File: sub-242_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-242
Checking File: sub-242_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-242
Checking File: sub-242_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-242
Checking File: sub-242_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-242
Checking File: sub-242_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-242
Checking File: sub-245_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-245
Checking File: sub-245_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-245
Checking File: sub-245_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-245
Checking File: sub-245_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-245
Checking File: sub-245_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-245
Checking File: sub-245_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-245
Checking File: sub-246_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-246
Checking File: sub-246_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-246
Checking File: sub-246_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-246
Checking File: sub-246_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-246
Checking File: sub-246_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-246
Checking File: sub-246_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-246
```

```
Checking File: sub-248_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-248
Checking File: sub-248_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-248
Checking File: sub-248_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-248
Checking File: sub-248_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-248
Checking File: sub-248_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-248
Checking File: sub-248_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-248
Checking File: sub-250_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-250
Checking File: sub-250_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-250
Checking File: sub-250_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-250
Checking File: sub-250_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-250
Checking File: sub-250_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-250
Checking File: sub-250_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-250
Checking File: sub-251_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-251
Checking File: sub-251_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-251
Checking File: sub-251_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-251
Checking File: sub-251_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-251
Checking File: sub-251_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-251
Checking File: sub-251_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-251
Checking File: sub-253_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-253
Checking File: sub-253_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-253
Checking File: sub-253_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-253
Checking File: sub-253_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-253
Checking File: sub-253_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-253
Checking File: sub-253_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-253
Checking File: sub-254_task-Conj19Sel_acq-b_bold.nii
```

```
Extracted Subject ID: sub-254
Checking File: sub-254_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-254
Checking File: sub-254_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-254
Checking File: sub-254_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-254
Checking File: sub-254_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-254
Checking File: sub-254_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-254
Checking File: sub-256_task-Conj19Sel_acq-a_bold.nii
Extracted Subject ID: sub-256
Checking File: sub-256_task-Conj1Inh_acq-a_bold.nii
Extracted Subject ID: sub-256
Checking File: sub-256_task-Conj9Inh_acq-a_bold.nii
Extracted Subject ID: sub-256
Checking File: sub-256_task-Feat19Sel_acq-b_bold.nii
Extracted Subject ID: sub-256
Checking File: sub-256_task-Feat1Inh_acq-b_bold.nii
Extracted Subject ID: sub-256
Checking File: sub-256_task-Feat9Inh_acq-b_bold.nii
Extracted Subject ID: sub-256
Checking File: sub-258_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-258
Checking File: sub-258_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-258
Checking File: sub-258_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-258
Checking File: sub-258_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-258
Checking File: sub-258_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-258
Checking File: sub-258_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-258
Checking File: sub-259_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-259
Checking File: sub-259_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-259
Checking File: sub-259_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-259
Checking File: sub-259_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-259
Checking File: sub-259_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-259
Checking File: sub-259_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-259
Checking File: sub-260_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-260
```

```
Checking File: sub-260_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-260
Checking File: sub-260_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-260
Checking File: sub-260_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-260
Checking File: sub-260_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-260
Checking File: sub-260_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-260
Checking File: sub-262_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-262
Checking File: sub-262_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-262
Checking File: sub-262_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-262
Checking File: sub-262_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-262
Checking File: sub-262_task-Feat1Inh_acq-b_bold.nii
Extracted Subject ID: sub-262
Checking File: sub-262_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-262
Checking File: sub-268_task-Conj19Sel_acq-b_bold.nii
Extracted Subject ID: sub-268
Checking File: sub-268_task-Conj1Inh_acq-b_bold.nii
Extracted Subject ID: sub-268
Checking File: sub-268_task-Conj9Inh_acq-b_bold.nii
Extracted Subject ID: sub-268
Checking File: sub-268_task-Feat19Sel_acq-a_bold.nii
Extracted Subject ID: sub-268
Checking File: sub-268_task-Feat1Inh_acq-a_bold.nii
Extracted Subject ID: sub-268
Checking File: sub-268_task-Feat9Inh_acq-a_bold.nii
Extracted Subject ID: sub-268
Final fMRI Data:
     subject_id
0       sub-213
1       sub-213
2       sub-213
3       sub-213
4       sub-213
..          ...
218     sub-268
219     sub-268
220     sub-268
221     sub-268
222     sub-268

[223 rows x 1 columns]
```

```
print("Task Data Sample:\n", task_data.head())
print("\nTask Data Shape:", task_data.shape)

print("\nfMRI Data Sample:\n", fmri_data.head())
print("\nfMRI Data Shape:", fmri_data.shape)
```

```
Task Data Sample:
     onset  duration  block block_type_intended
block_type_performed  \
0    0.00     38.69      1              single               single
1   38.69     38.87      2               array                array
2   77.56     38.69      3              single               single
3  116.25     38.87      4               array                array
4  155.12     38.69      5              single               single

   response_time_avg  correct_total errors_total  false-neg  false-pos
NR  \
0             0.5679           17.0            1        NaN        1.0
NaN
1             0.6947           18.0            0        NaN        NaN
NaN
2             0.5872           18.0            0        NaN        NaN
NaN
3             0.7014           18.0            0        NaN        NaN
NaN
4             0.5265           18.0            0        NaN        NaN
NaN

  comments subject_id  false-no_go  false-go
0      NaN    sub-213          NaN       NaN
1      NaN    sub-213          NaN       NaN
2      NaN    sub-213          NaN       NaN
3      NaN    sub-213          NaN       NaN
4      NaN    sub-213          NaN       NaN

Task Data Shape: (2753, 15)

fMRI Data Sample:
   subject_id
0    sub-213
1    sub-213
2    sub-213
3    sub-213
4    sub-213

fMRI Data Shape: (223, 1)
```

```
print("Task Data Columns:", task_data.columns)
print("fMRI Data Columns:", fmri_data.columns)
```

```
Task Data Columns: Index(['onset', 'duration', 'block',
'block_type_intended',
       'block_type_performed', 'response_time_avg', 'correct_total',
       'errors_total', 'false-neg', 'false-pos', 'NR', 'comments',
       'subject_id', 'false-no_go', 'false-go'],
      dtype='object')
fMRI Data Columns: Index(['subject_id'], dtype='object')

final_data = pd.merge(task_data, fmri_data, on="subject_id",
how="inner")
print("Final Merged Data Sample:\n", final_data.head())
print("Final Data Shape:", final_data.shape)

Final Merged Data Sample:
    onset  duration  block block_type_intended block_type_performed  \
0    0.0     38.69      1              single               single
1    0.0     38.69      1              single               single
2    0.0     38.69      1              single               single
3    0.0     38.69      1              single               single
4    0.0     38.69      1              single               single

   response_time_avg  correct_total errors_total  false-neg  false-pos
NR  \
0             0.5679           17.0            1        NaN        1.0
NaN
1             0.5679           17.0            1        NaN        1.0
NaN
2             0.5679           17.0            1        NaN        1.0
NaN
3             0.5679           17.0            1        NaN        1.0
NaN
4             0.5679           17.0            1        NaN        1.0
NaN

  comments subject_id  false-no_go  false-go
0      NaN    sub-213          NaN       NaN
1      NaN    sub-213          NaN       NaN
2      NaN    sub-213          NaN       NaN
3      NaN    sub-213          NaN       NaN
4      NaN    sub-213          NaN       NaN
Final Data Shape: (16311, 15)

from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

categorical_columns =
final_data.select_dtypes(include=['object']).columns.tolist()
print("Categorical Columns:", categorical_columns)

final_data_encoded = pd.get_dummies(final_data,
```

```
columns=categorical_columns, drop_first=True)

X = final_data_encoded.drop(columns=["response_time_avg",
"subject_id"], errors="ignore")
y = final_data_encoded["response_time_avg"]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y,
test_size=0.2, random_state=42)

print("Data prepared for Neural Network!")
print("Training Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)
```

```
Categorical Columns: ['block_type_intended', 'block_type_performed',
'errors_total', 'comments', 'subject_id']
Data prepared for Neural Network!
Training Data Shape: (13048, 160)
Testing Data Shape: (3263, 160)
```

```
import numpy as np
import pandas as pd

print("NaN in X_train:", np.isnan(X_train).sum())
print("NaN in y_train:", np.isnan(y_train).sum())

print("Inf in X_train:", np.isinf(X_train).sum())
print("Inf in y_train:", np.isinf(y_train).sum())

print(pd.DataFrame(X_train).describe())

print(final_data.dtypes)

X_train = np.nan_to_num(X_train)
y_train = np.nan_to_num(y_train)
```

```
NaN in X_train: 58526
NaN in y_train: 359
Inf in X_train: 0
Inf in y_train: 0
                0             1             2             3
4    \
count  13048.000000  13048.000000  13048.000000  12641.000000
1510.000000
mean       0.003505     -0.001233      0.003562     -0.003946    -
0.001858
std        1.000695      1.005361      1.000686      1.003064
1.008523
min       -1.605193     -6.596402     -1.603733     -7.214100    -
```

```
0.591077
25%       -0.737815      0.201159     -0.733466      0.064610    -
0.591077
50%        0.129562      0.201159      0.136801      0.519530    -
0.591077
75%        0.996940      0.218723      1.007069      0.519530
1.006269
max        1.782265      1.026702      1.587247      0.519530
4.200962

                5           6            7            8             9
...  \
count  1296.000000  524.000000  1773.000000  2018.000000  13048.000000
...
mean     -0.009582    0.011988     0.003012    -0.005506      0.002609
...
std       0.991801    1.024469     1.010411     0.997839      1.000945
...
min      -0.527926   -0.859283    -0.541346    -0.625488     -0.710230
...
25%      -0.527926   -0.859283    -0.541346    -0.625488     -0.710230
...
50%      -0.527926    0.207413    -0.541346    -0.625488     -0.710230
...
75%       0.860036    0.207413     0.837435     0.303334      1.407994
...
max       5.023924    4.474197     6.352559     4.018620      1.407994
...

                150           151           152           153
154  \
count  13048.000000  13048.000000  13048.000000  13048.000000
13048.000000
mean       0.004021     -0.002663      0.001633     -0.001231     -
0.002185
std        1.009934      0.992151      1.004841      0.996401
0.993570
min       -0.194405     -0.164942     -0.164942     -0.164942     -
0.164942
25%       -0.194405     -0.164942     -0.164942     -0.164942     -
0.164942
50%       -0.194405     -0.164942     -0.164942     -0.164942     -
0.164942
75%       -0.194405     -0.164942     -0.164942     -0.164942     -
0.164942
max        5.143889      6.062751      6.062751      6.062751
6.062751

                155           156           157           158
159
```

```
count  13048.000000  13048.000000  13048.000000  13048.000000
13048.000000
mean       0.005739     -0.000276      0.000201      0.000185
0.001033
std        1.016208      0.999223      1.000631      1.000531
1.002992
min       -0.170734     -0.164942     -0.164942     -0.181827     -
0.169590
25%       -0.170734     -0.164942     -0.164942     -0.181827     -
0.169590
50%       -0.170734     -0.164942     -0.164942     -0.181827     -
0.169590
75%       -0.170734     -0.164942     -0.164942     -0.181827     -
0.169590
max        5.857064      6.062751      6.062751      5.499739
5.896587

[8 rows x 160 columns]
onset                   float64
duration                float64
block                     int64
block_type_intended      object
block_type_performed     object
response_time_avg       float64
correct_total           float64
errors_total             object
false-neg               float64
false-pos               float64
NR                      float64
comments                 object
subject_id               object
false-no_go             float64
false-go                float64
dtype: object
```

```
X_train = np.nan_to_num(X_train, nan=np.nanmean(X_train))
y_train = np.nan_to_num(y_train, nan=np.nanmean(y_train))

numeric_cols = final_data.select_dtypes(include=["number"]).columns
final_data[numeric_cols] =
final_data[numeric_cols].fillna(final_data[numeric_cols].mean())

print(final_data.dtypes)
```

```
onset                   float64
duration                float64
block                     int64
block_type_intended         int8
block_type_performed        int8
response_time_avg       float64
```

```
correct_total            float64
errors_total             float64
false-neg                float64
false-pos                float64
NR                       float64
comments                   int8
subject_id                object
false-no_go              float64
false-go                 float64
dtype: object
```

```
numeric_cols = final_data.select_dtypes(include=["number"]).columns

final_data[numeric_cols] =
final_data[numeric_cols].fillna(final_data[numeric_cols].mean())

print("NaN values handled successfully!")
```

```
NaN values handled successfully!
```

```
print(final_data.isna().sum())
```

```
onset                    0
duration                 0
block                    0
block_type_intended      0
block_type_performed     0
response_time_avg        0
correct_total            0
errors_total             0
false-neg                0
false-pos                0
NR                       0
comments                 0
subject_id               0
false-no_go              0
false-go                 0
dtype: int64
```

```
print(final_data.columns)
```

```
Index(['onset', 'duration', 'block', 'block_type_intended',
       'block_type_performed', 'response_time_avg', 'correct_total',
       'errors_total', 'false-neg', 'false-pos', 'NR', 'comments',
       'subject_id', 'false-no_go', 'false-go'],
      dtype='object')
```

```
from sklearn.preprocessing import StandardScaler

X = final_data.drop(columns=["response_time_avg", "subject_id"])
```

```
y = final_data["response_time_avg"]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print("Features scaled successfully!")
```

```
Features scaled successfully!
```

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y,
test_size=0.2, random_state=42)

print(f"Training Samples: {len(X_train)}, Testing Samples:
{len(X_test)}")
```

```
Training Samples: 13048, Testing Samples: 3263
```

```
from sklearn.preprocessing import StandardScaler

X = final_data.drop(columns=["response_time_avg", "subject_id"])
y = final_data["response_time_avg"]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

print(f"NaN in X_scaled: {np.isnan(X_scaled).sum()}")
print(f"NaN in y: {np.isnan(y).sum()}")

y = np.array(y)
```

```
NaN in X_scaled: 0
NaN in y: 0
```

```
from sklearn.model_selection import train_test_split
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout
from tensorflow.keras.optimizers import Adam

X_train, X_test, y_train, y_test = train_test_split(X_scaled, y,
test_size=0.2, random_state=42)

model = Sequential([
    Dense(128, activation='relu', input_shape=(X_train.shape[1],)),
    Dropout(0.3),
    Dense(64, activation='relu'),
    Dropout(0.3),
    Dense(1, activation='linear')
])

model.compile(optimizer=Adam(learning_rate=0.001),
```

```
loss='mean_squared_error', metrics=['mae'])

history = model.fit(X_train, y_train, validation_data=(X_test,
y_test), epochs=50, batch_size=16)

loss, mae = model.evaluate(X_test, y_test)
print(f"Test Loss: {loss}")
print(f"Test MAE: {mae}")
```

```
Epoch 1/50

D:\Anaconda\Lib\site-packages\keras\src\layers\core\dense.py:87:
UserWarning: Do not pass an `input_shape`/`input_dim` argument to a
layer. When using Sequential models, prefer using an `Input(shape)`
object as the first layer in the model instead.
  super().__init__(activity_regularizer=activity_regularizer,
**kwargs)

816/816 ━━━━━━━━━━━━━━━━━━━━ 2s 1ms/step - loss: 0.1204 - mae: 0.2385
- val_loss: 0.0293 - val_mae: 0.1337
Epoch 2/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0372 - mae: 0.1524
- val_loss: 0.0271 - val_mae: 0.1293
Epoch 3/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0308 - mae: 0.1402
- val_loss: 0.0269 - val_mae: 0.1287
Epoch 4/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0294 - mae: 0.1376
- val_loss: 0.0251 - val_mae: 0.1251
Epoch 5/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0272 - mae: 0.1320
- val_loss: 0.0252 - val_mae: 0.1258
Epoch 6/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0273 - mae: 0.1324
- val_loss: 0.0245 - val_mae: 0.1240
Epoch 7/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0268 - mae: 0.1310
- val_loss: 0.0243 - val_mae: 0.1244
Epoch 8/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0264 - mae: 0.1293
- val_loss: 0.0242 - val_mae: 0.1237
Epoch 9/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0263 - mae: 0.1294
- val_loss: 0.0239 - val_mae: 0.1217
Epoch 10/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0266 - mae: 0.1298
- val_loss: 0.0239 - val_mae: 0.1232
Epoch 11/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0262 - mae: 0.1294
- val_loss: 0.0236 - val_mae: 0.1225
```

```
Epoch 12/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0252 - mae: 0.1267
- val_loss: 0.0240 - val_mae: 0.1223
Epoch 13/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0264 - mae: 0.1291
- val_loss: 0.0238 - val_mae: 0.1235
Epoch 14/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0258 - mae: 0.1278
- val_loss: 0.0236 - val_mae: 0.1217
Epoch 15/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0253 - mae: 0.1261
- val_loss: 0.0236 - val_mae: 0.1214
Epoch 16/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0256 - mae: 0.1272
- val_loss: 0.0239 - val_mae: 0.1228
Epoch 17/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0252 - mae: 0.1262
- val_loss: 0.0233 - val_mae: 0.1210
Epoch 18/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0253 - mae: 0.1266
- val_loss: 0.0234 - val_mae: 0.1214
Epoch 19/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0259 - mae: 0.1287
- val_loss: 0.0231 - val_mae: 0.1200
Epoch 20/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0245 - mae: 0.1248
- val_loss: 0.0238 - val_mae: 0.1220
Epoch 21/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0250 - mae: 0.1254
- val_loss: 0.0232 - val_mae: 0.1188
Epoch 22/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0253 - mae: 0.1263
- val_loss: 0.0234 - val_mae: 0.1217
Epoch 23/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0249 - mae: 0.1250
- val_loss: 0.0233 - val_mae: 0.1205
Epoch 24/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0249 - mae: 0.1251
- val_loss: 0.0234 - val_mae: 0.1219
Epoch 25/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0248 - mae: 0.1250
- val_loss: 0.0232 - val_mae: 0.1200
Epoch 26/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0253 - mae: 0.1259
- val_loss: 0.0238 - val_mae: 0.1219
Epoch 27/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0253 - mae: 0.1269
- val_loss: 0.0230 - val_mae: 0.1202
Epoch 28/50
```

```
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0248 - mae: 0.1254
- val_loss: 0.0228 - val_mae: 0.1185
Epoch 29/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0243 - mae: 0.1240
- val_loss: 0.0231 - val_mae: 0.1210
Epoch 30/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0254 - mae: 0.1265
- val_loss: 0.0233 - val_mae: 0.1191
Epoch 31/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0248 - mae: 0.1247
- val_loss: 0.0232 - val_mae: 0.1205
Epoch 32/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0247 - mae: 0.1244
- val_loss: 0.0229 - val_mae: 0.1187
Epoch 33/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0247 - mae: 0.1242
- val_loss: 0.0229 - val_mae: 0.1188
Epoch 34/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0246 - mae: 0.1244
- val_loss: 0.0229 - val_mae: 0.1195
Epoch 35/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0241 - mae: 0.1229
- val_loss: 0.0229 - val_mae: 0.1202
Epoch 36/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0246 - mae: 0.1250
- val_loss: 0.0228 - val_mae: 0.1177
Epoch 37/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0248 - mae: 0.1245
- val_loss: 0.0227 - val_mae: 0.1186
Epoch 38/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0247 - mae: 0.1240
- val_loss: 0.0228 - val_mae: 0.1184
Epoch 39/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0245 - mae: 0.1240
- val_loss: 0.0231 - val_mae: 0.1206
Epoch 40/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0241 - mae: 0.1229
- val_loss: 0.0230 - val_mae: 0.1192
Epoch 41/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0245 - mae: 0.1240
- val_loss: 0.0231 - val_mae: 0.1215
Epoch 42/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0245 - mae: 0.1237
- val_loss: 0.0236 - val_mae: 0.1223
Epoch 43/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0242 - mae: 0.1231
- val_loss: 0.0230 - val_mae: 0.1193
Epoch 44/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0241 - mae: 0.1224
```

```
- val_loss: 0.0224 - val_mae: 0.1172
Epoch 45/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0243 - mae: 0.1233
- val_loss: 0.0225 - val_mae: 0.1178
Epoch 46/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0242 - mae: 0.1227
- val_loss: 0.0228 - val_mae: 0.1186
Epoch 47/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0245 - mae: 0.1238
- val_loss: 0.0231 - val_mae: 0.1184
Epoch 48/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0243 - mae: 0.1233
- val_loss: 0.0222 - val_mae: 0.1167
Epoch 49/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0237 - mae: 0.1213
- val_loss: 0.0229 - val_mae: 0.1188
Epoch 50/50
816/816 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - loss: 0.0242 - mae: 0.1231
- val_loss: 0.0229 - val_mae: 0.1194
102/102 ━━━━━━━━━━━━━━━━━━━━ 0s 895us/step - loss: 0.0230 - mae:
0.1193
Test Loss: 0.022939201444387436
Test MAE: 0.11935841292142868
```

```
import joblib
from tensorflow.keras.models import load_model

model.save("task_performance_model.h5")

joblib.dump(scaler, "scaler.pkl")

final_data.to_csv("final_task_performance_data.csv", index=False)

print(" Model and dataset saved successfully!")
```

```
WARNING:absl:You are saving your model as an HDF5 file via
`model.save()` or `keras.saving.save_model(model)`. This file format
is considered legacy. We recommend using instead the native Keras
format, e.g. `model.save('my_model.keras')` or
`keras.saving.save_model(model, 'my_model.keras')`.

 Model and dataset saved successfully!
```

```
import matplotlib.pyplot as plt
from sklearn.decomposition import PCA

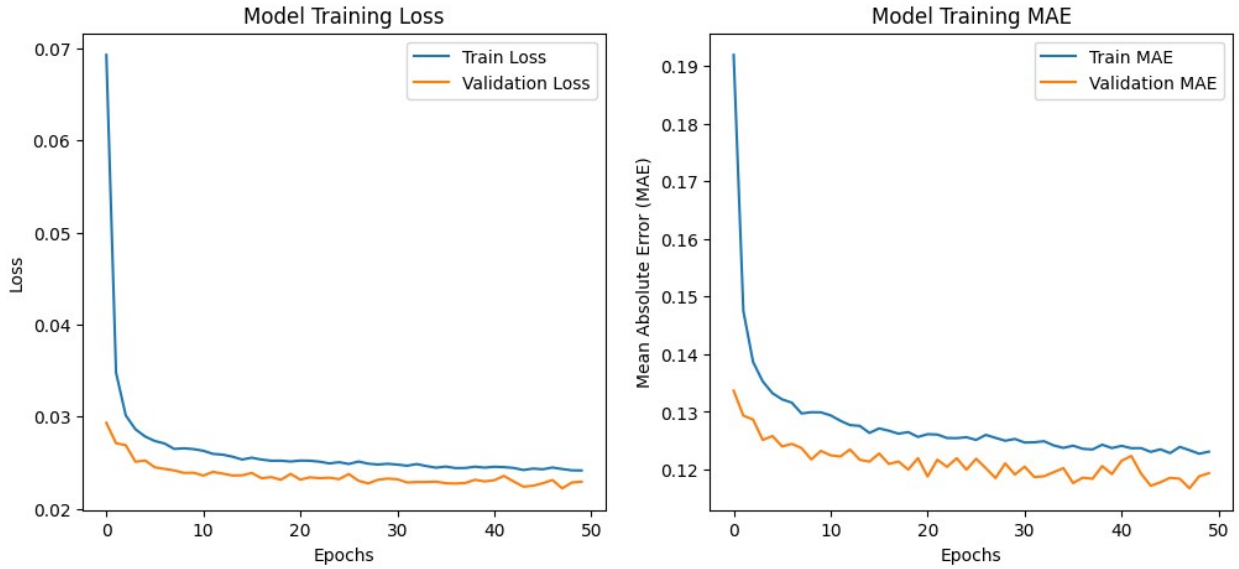
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
```

```
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.title('Model Training Loss')
plt.legend()

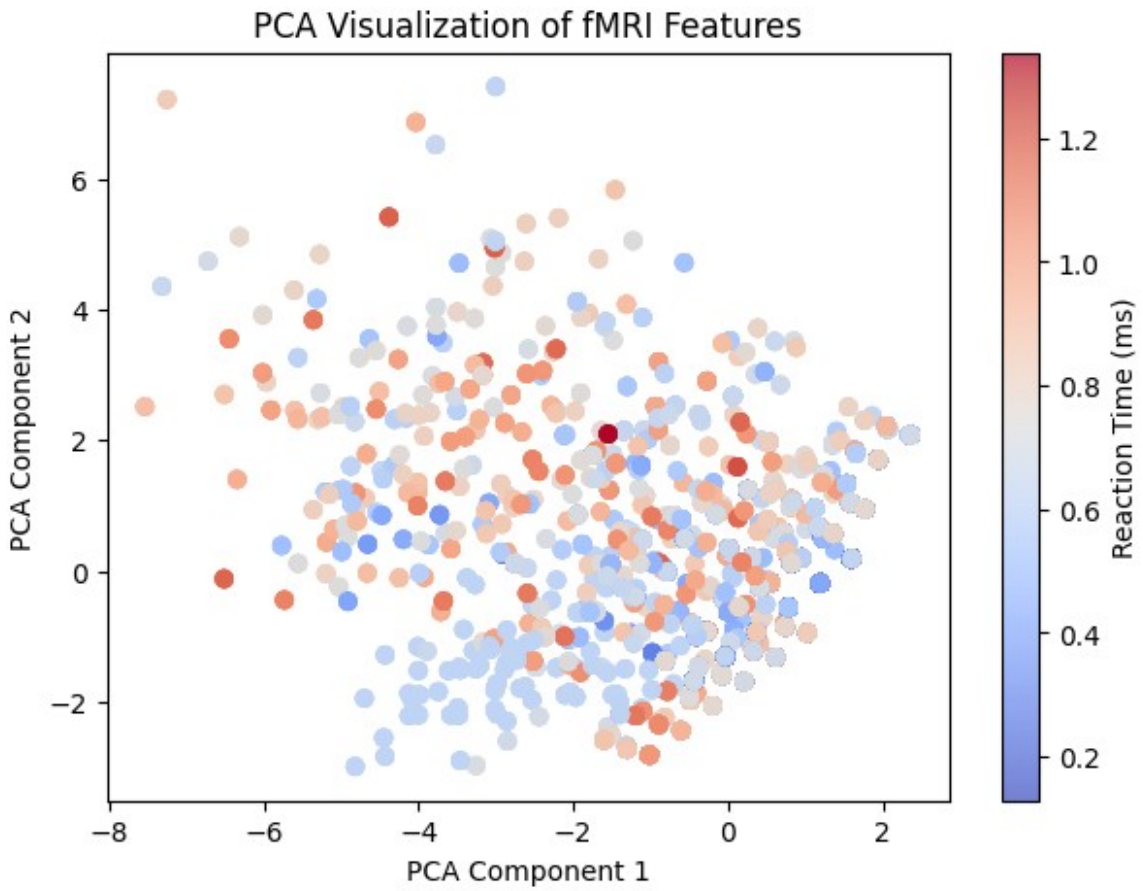
plt.subplot(1, 2, 2)
plt.plot(history.history['mae'], label='Train MAE')
plt.plot(history.history['val_mae'], label='Validation MAE')
plt.xlabel('Epochs')
plt.ylabel('Mean Absolute Error (MAE)')
plt.title('Model Training MAE')
plt.legend()

plt.show()
```




```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

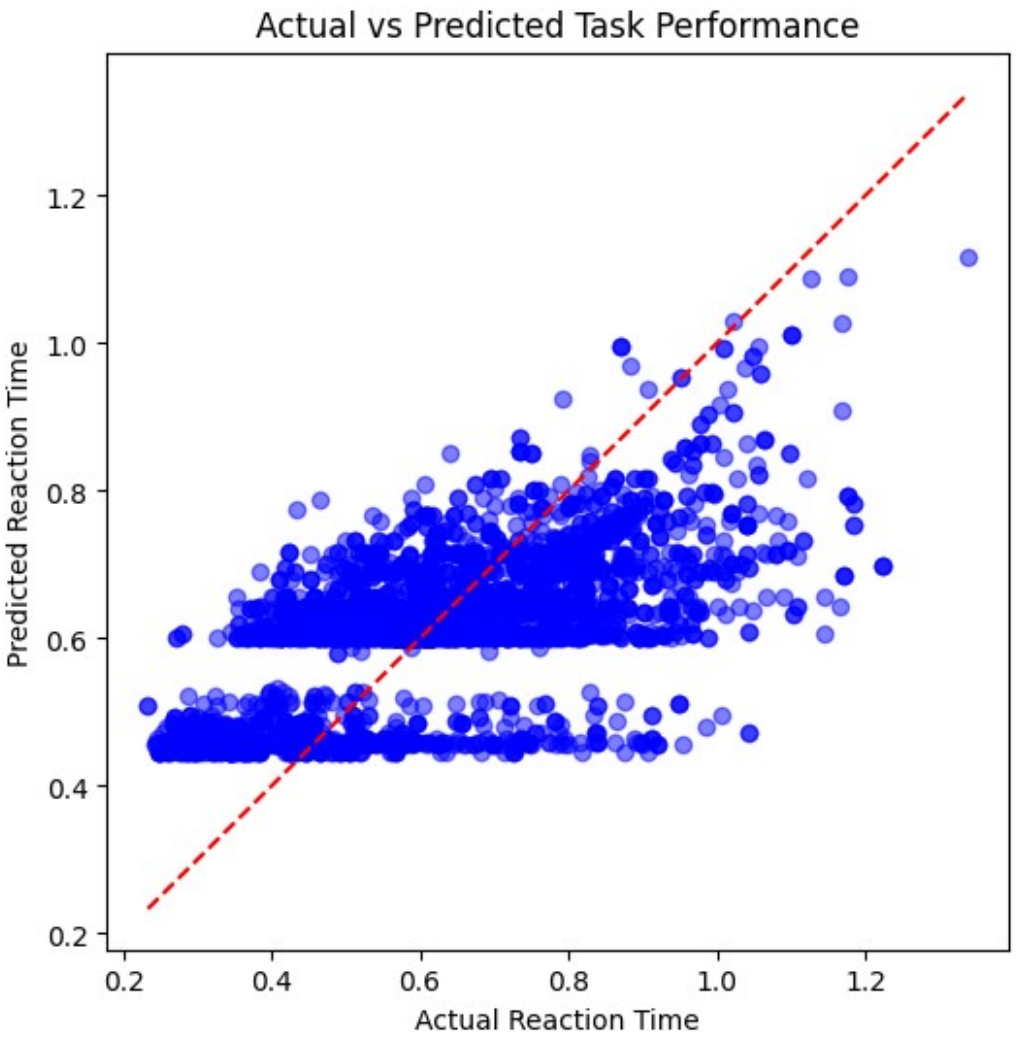
plt.figure(figsize=(7, 5))
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y, cmap='coolwarm', alpha=0.7)
plt.colorbar(label="Reaction Time (ms)")
plt.xlabel('PCA Component 1')
plt.ylabel('PCA Component 2')
plt.title('PCA Visualization of fMRI Features')
plt.show()
```




```
y_pred = model.predict(X_test)

plt.figure(figsize=(6, 6))
plt.scatter(y_test, y_pred, alpha=0.5, color='blue')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()],
'r--')
plt.xlabel('Actual Reaction Time')
plt.ylabel('Predicted Reaction Time')
plt.title('Actual vs Predicted Task Performance')
plt.show()
```

```
102/102 ━━━━━━━━━━━━━━━━━━━━ 0s 985us/step
```



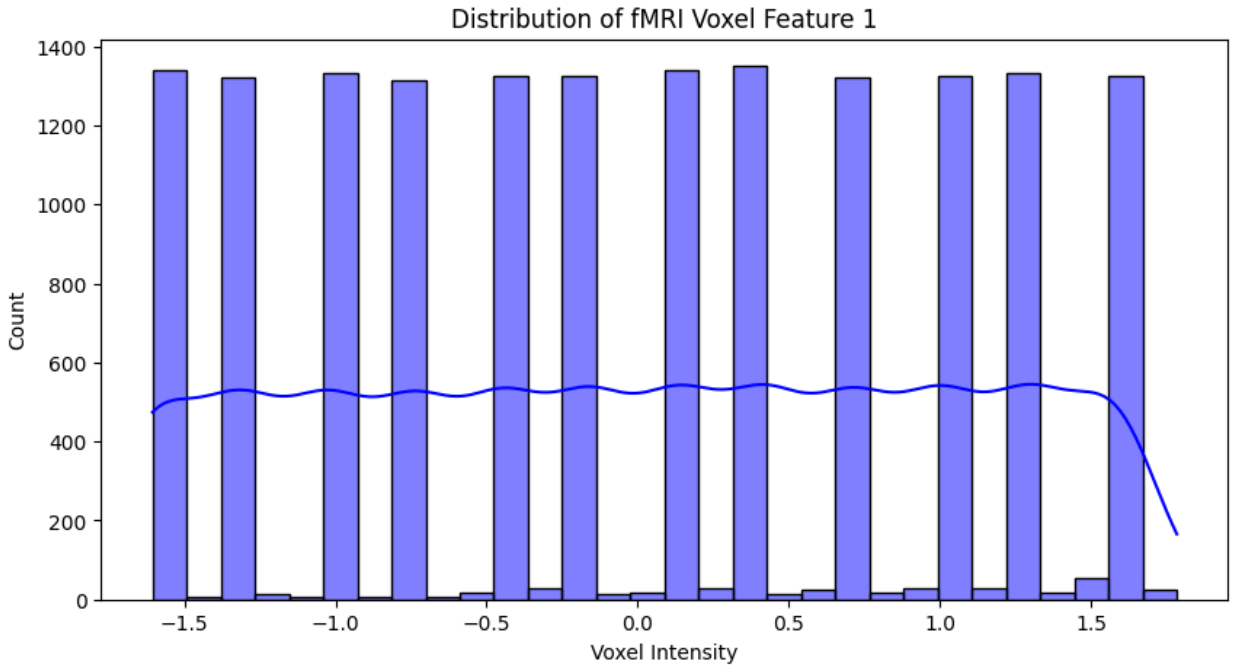

```
!pip install shap

Collecting shap
  Downloading shap-0.47.1-cp312-cp312-win_amd64.whl.metadata (25 kB)
Requirement already satisfied: numpy in d:\anaconda\lib\site-packages
(from shap) (1.26.4)
Requirement already satisfied: scipy in d:\anaconda\lib\site-packages
(from shap) (1.13.1)
Requirement already satisfied: scikit-learn in d:\anaconda\lib\site-
packages (from shap) (1.5.1)
Requirement already satisfied: pandas in d:\anaconda\lib\site-packages
(from shap) (2.2.2)
Requirement already satisfied: tqdm>=4.27.0 in d:\anaconda\lib\site-
packages (from shap) (4.66.5)
Requirement already satisfied: packaging>20.9 in d:\anaconda\lib\site-
packages (from shap) (24.1)
Collecting slicer==0.0.8 (from shap)
  Downloading slicer-0.0.8-py3-none-any.whl.metadata (4.0 kB)
```

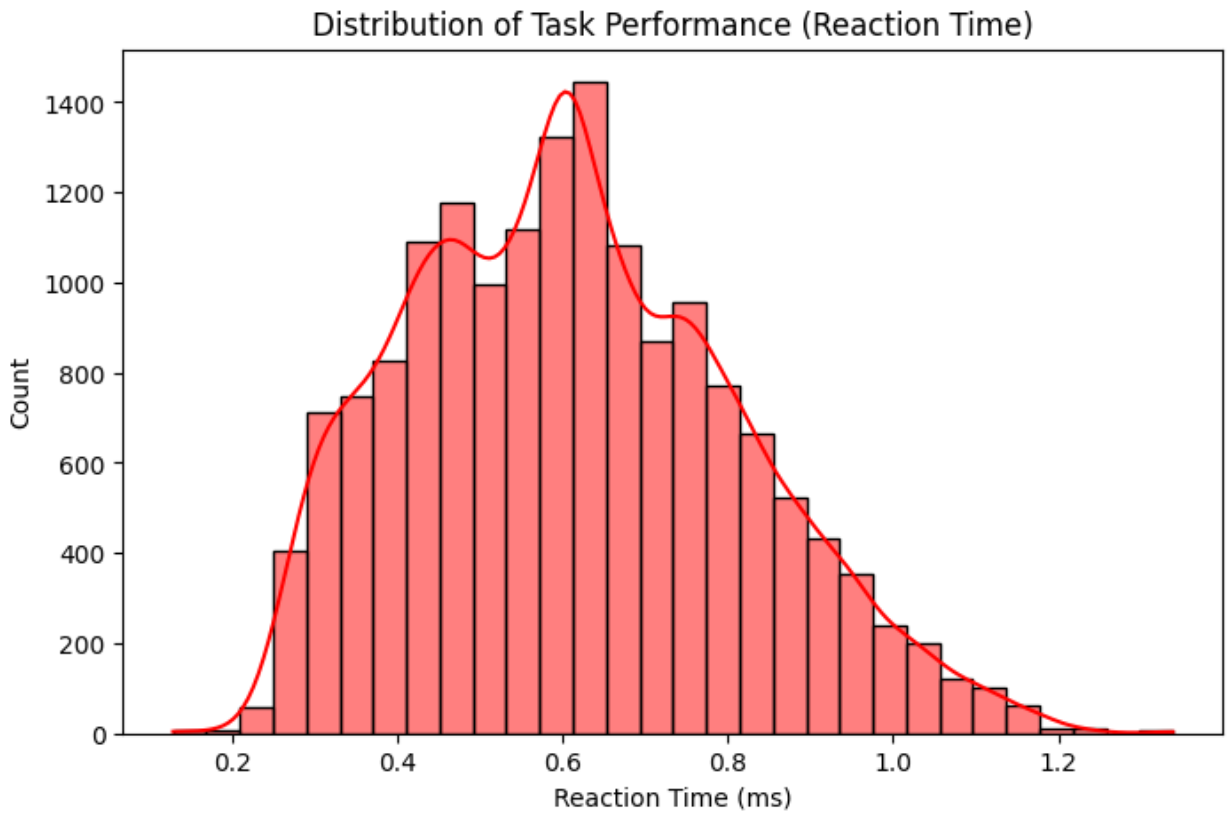
```
Requirement already satisfied: numba>=0.54 in d:\anaconda\lib\site-
packages (from shap) (0.60.0)
Requirement already satisfied: cloudpickle in d:\anaconda\lib\site-
packages (from shap) (3.0.0)
Requirement already satisfied: typing-extensions in d:\anaconda\lib\
site-packages (from shap) (4.11.0)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in d:\
anaconda\lib\site-packages (from numba>=0.54->shap) (0.43.0)
Requirement already satisfied: colorama in d:\anaconda\lib\site-
packages (from tqdm>=4.27.0->shap) (0.4.6)
Requirement already satisfied: python-dateutil>=2.8.2 in d:\anaconda\
lib\site-packages (from pandas->shap) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in d:\anaconda\lib\site-
packages (from pandas->shap) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in d:\anaconda\lib\site-
packages (from pandas->shap) (2023.3)
Requirement already satisfied: joblib>=1.2.0 in d:\anaconda\lib\site-
packages (from scikit-learn->shap) (1.4.2)
Requirement already satisfied: threadpoolctl>=3.1.0 in d:\anaconda\
lib\site-packages (from scikit-learn->shap) (3.5.0)
Requirement already satisfied: six>=1.5 in d:\anaconda\lib\site-
packages (from python-dateutil>=2.8.2->pandas->shap) (1.16.0)
Downloading shap-0.47.1-cp312-cp312-win_amd64.whl (490 kB)
Downloading slicer-0.0.8-py3-none-any.whl (15 kB)
Installing collected packages: slicer, shap
Successfully installed shap-0.47.1 slicer-0.0.8
```

```python
import seaborn as sns
import shap
import numpy as np

plt.figure(figsize=(10, 5))
sns.histplot(X_scaled[:, 0], kde=True, bins=30, color="blue")
plt.xlabel("Voxel Intensity")
plt.ylabel("Count")
plt.title("Distribution of fMRI Voxel Feature 1")
plt.show()
```




```
plt.figure(figsize=(8, 5))
sns.histplot(y, kde=True, bins=30, color="red")
plt.xlabel("Reaction Time (ms)")
plt.ylabel("Count")
plt.title("Distribution of Task Performance (Reaction Time)")
plt.show()
```

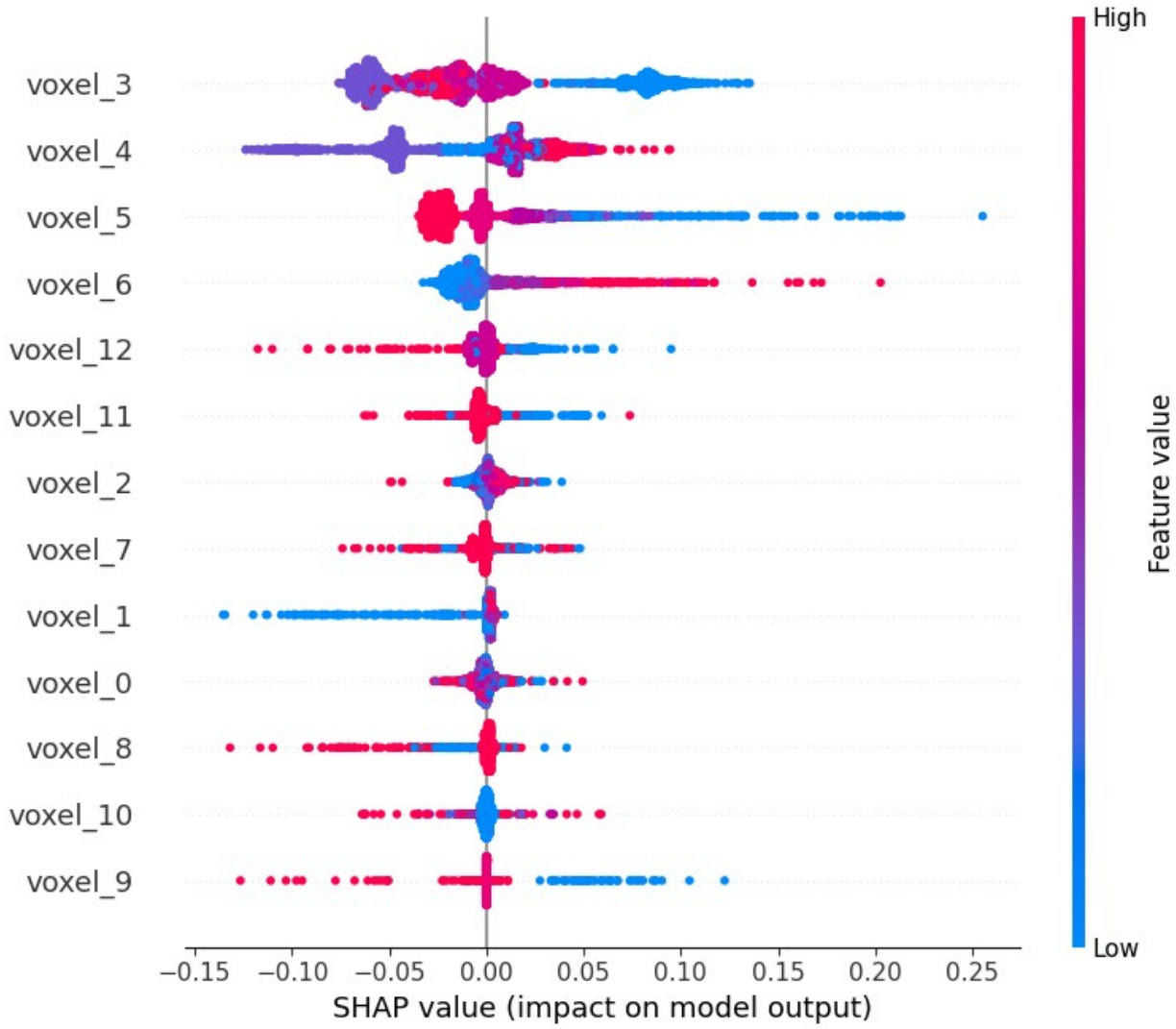



```
explainer = shap.Explainer(model, X_test)
shap_values = explainer(X_test)

plt.figure(figsize=(10, 5))
shap.summary_plot(shap_values, X_test, feature_names=[f"voxel_{i}" for
i in range(X_test.shape[1])])
```

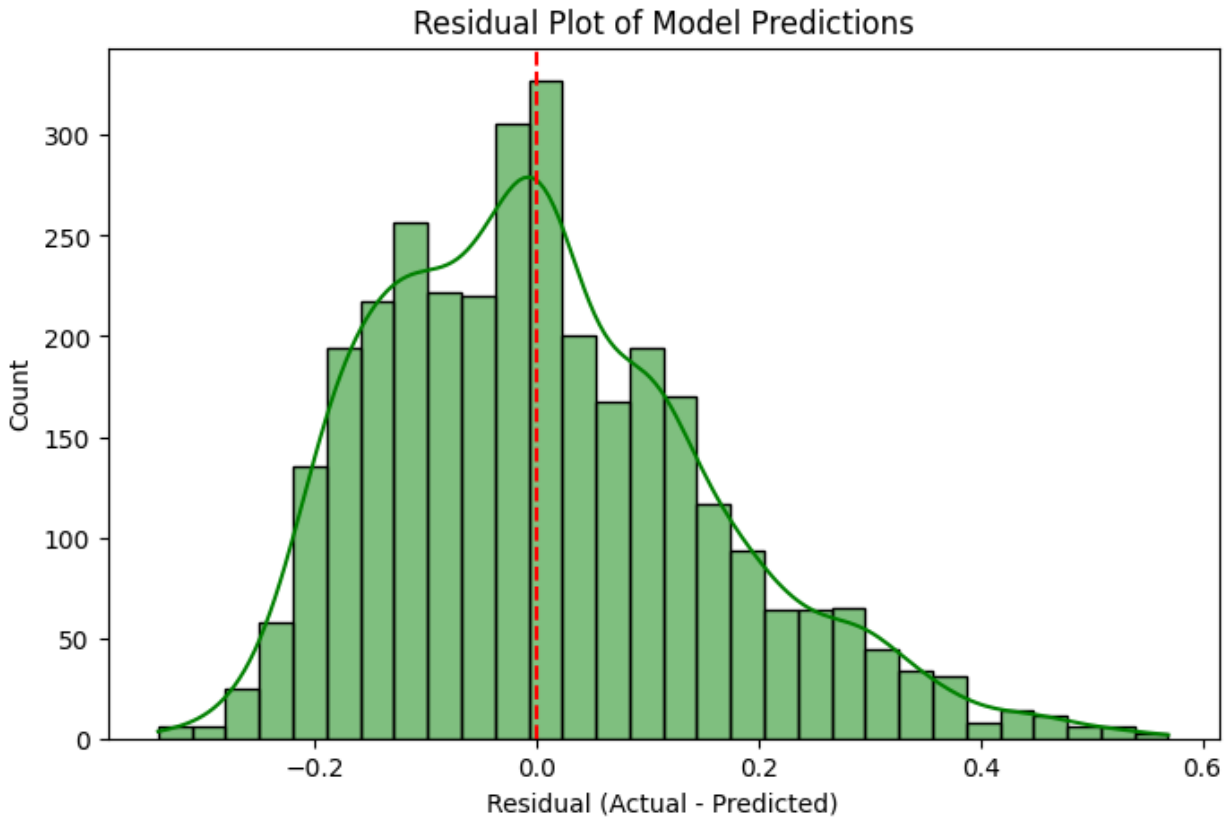
```
PermutationExplainer explainer: 3264it [03:42, 14.28it/s]

C:\Users\Admin\AppData\Local\Temp\ipykernel_8324\575048666.py:5:
FutureWarning: The NumPy global RNG was seeded by calling
`np.random.seed`. In a future version this function will no longer use
the global RNG. Pass `rng` explicitly to opt-in to the new behaviour
and silence this warning.
  shap.summary_plot(shap_values, X_test, feature_names=[f"voxel_{i}"
for i in range(X_test.shape[1])])
```




```
residuals = y_test - y_pred.flatten()

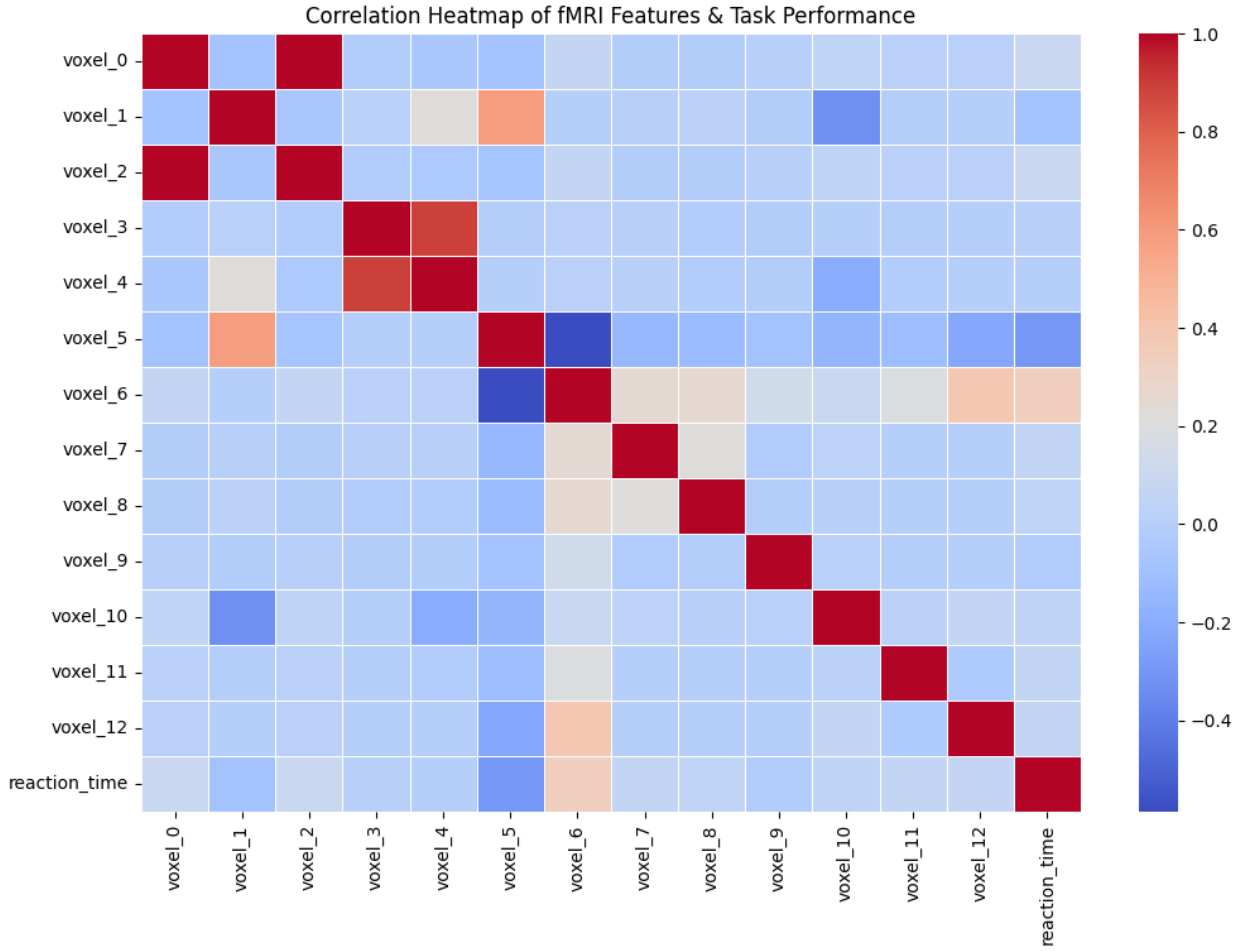
plt.figure(figsize=(8, 5))
sns.histplot(residuals, kde=True, bins=30, color="green")
plt.axvline(0, color='red', linestyle='--')
plt.xlabel("Residual (Actual - Predicted)")
plt.ylabel("Count")
plt.title("Residual Plot of Model Predictions")
plt.show()
```




```
df_combined = pd.DataFrame(X_scaled, columns=[f"voxel_{i}" for i in
range(X_scaled.shape[1])])
df_combined['reaction_time'] = y

corr_matrix = df_combined.corr()

plt.figure(figsize=(12, 8))
sns.heatmap(corr_matrix, cmap="coolwarm", annot=False, linewidths=0.5)
plt.title("Correlation Heatmap of fMRI Features & Task Performance")
plt.show()
```


Correlation Heatmap of fMRI Features & Task Performance
voxel_0
voxel_1
voxel_2
voxel_3
voxel_4
voxel_5
voxel_6
voxel_7
voxel_8
voxel_9
voxel_10
voxel_11
voxel_12
reaction_time
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4